



UIT2734
IOT ARCHITECTURES
AND APPLICATIONS

Unit 1 : Fundamentals of IoT

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Reference Books

1. David Hanes et al., "**IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things**" (Cisco Press, 2017)
2. Arshdeep Bahga and Vijay Madisetti, "**Internet of Things: A Hands-on Approach**"

What is IoT?

- IoT is an intelligent network of **smart physical objects**—devices, vehicles, buildings, and infrastructure—embedded with:
 - **Sensors / Actuators** (to sense and act)
 - **Computation Unit** (to process)
 - **Memory Unit** (to store)
 - **Power Source** (to sustain)
 - **Network Connectivity** (to communicate)
 - **Primary Capabilities:**
 - **Collect & Exchange** data across the digital-physical divide.
 - **Analyze** raw data to extract actionable insights.
 - **Respond** intelligently—autonomously or with human guidance.
 - **Goal of IoT:** “*To connect the unconnected.*”
- 



Traditional "Unconnected"	IoT-Enabled Smart Object
Standard streetlight	Smart streetlight that dims based on pedestrian presence
Industrial motor	Motor that self-reports vibration for predictive maintenance
Agricultural field	Soil sensor network that automates irrigation
Residential thermostat	Thermostat that learns occupancy patterns

Brief History of IoT

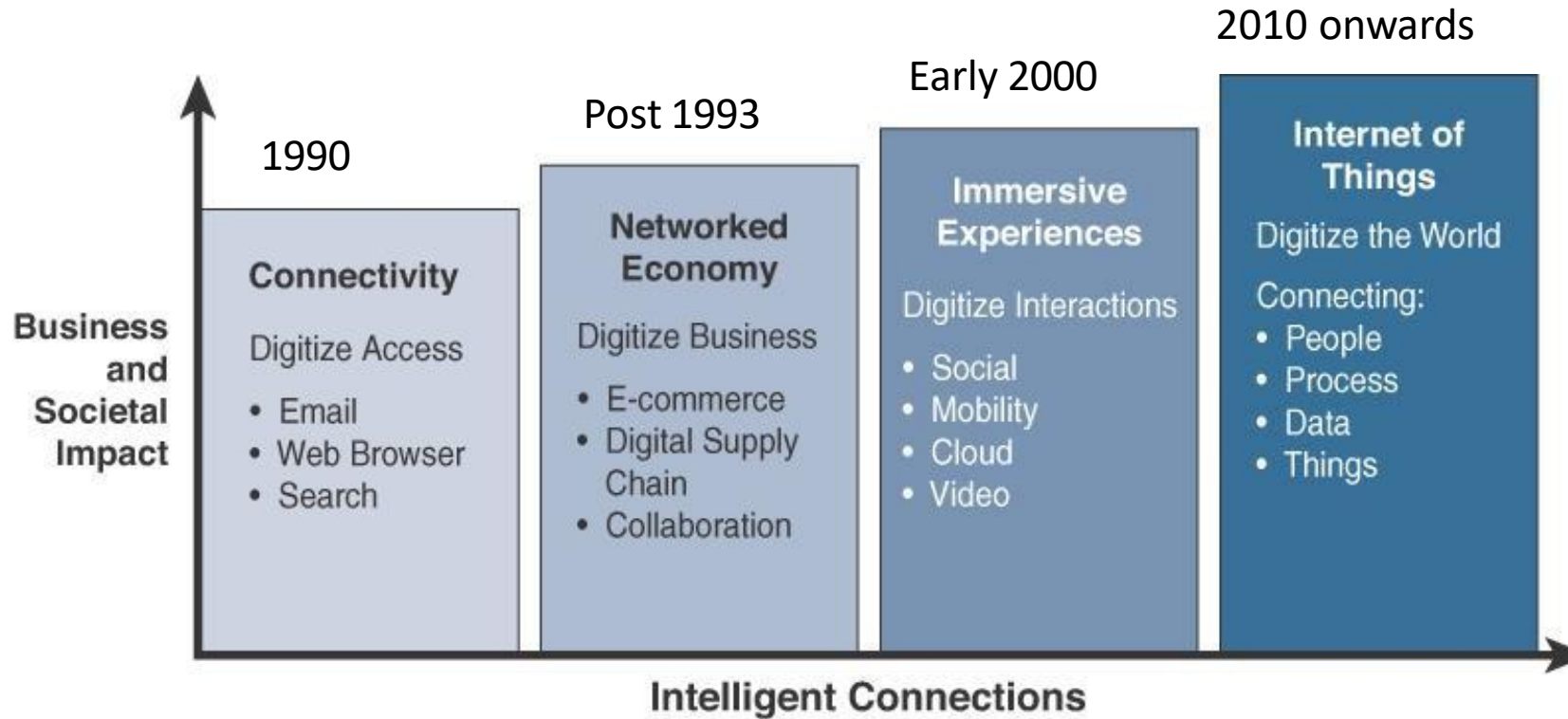
- The term "Internet of Things" was likely coined by Kevin Ashton of Procter & Gamble, later MIT's Auto-ID Center, in 1999.
 - Kevin Ashton said –
 - “In 20th century, computers were brains without senses — they only knew what we told them.”
 - “Now in 21st century, computers are sensing things for themselves!”
- In the early 1980s at Carnegie Mellon University, a group of students created a way to get their campus's Coca-Cola vending machine to report on its contents through a network.

Brief History of IoT...

- In 1990, John Romkey, developer of the first TCP/IP stack for IBM PC in 1983, connected a toaster to the internet for the first time.
- In 1991, a group of students at the University of Cambridge used a web camera to report on coffee available in their computer labs' coffee pot.
- At the **beginning of the 21st Century**, LG (Year 2000) Electronics introduced the world's first refrigerator connected to the internet

Brief History of IoT

- The popularity of IoT did not accelerate until 2010/2011 and reached mass market from 2013-14.
- Definition of the IoT has also evolved over time.



Evolutionary Phases of the Internet

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Evolution of the Internet (In Tabular)

Phase	Time Period	Theme & Focus	Key Drivers	Impact (Brief)
1. Connectivity	Mid-1990s	Getting Online	Dial-up, Email, University/Corporate Access	Made basic Internet access a reality for early adopters.
2. Networked Economy	Late 1990s – 2000s	Efficiency & Profit	E-commerce, Digital Supply Chains, Vendor-Producer Integration	Disrupted retail; tightly interlinked global economy.
3. Immersive Experiences	Late 2000s – 2010s	Pervasive Collaboration	Social Media, Smartphones, Multi-channel Communication	Digitized human interaction; ubiquitous global connectivity.
4. Internet of Things (IoT)	2010s – Present	Intelligent Automation	M2M, 99% Things Unconnected, Big Data Analytics	Drives automation, new insights, and process efficiencies.

Genesis of IoT

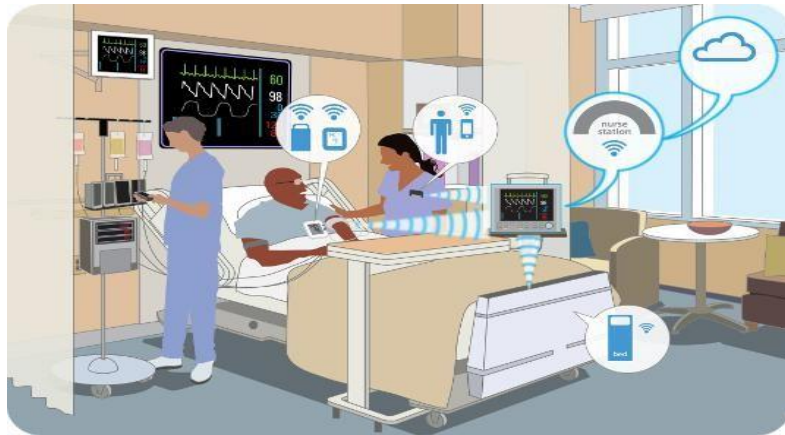
- IoT represents the evolution from M2M to a broader ecosystem
- **Historical Progression:**
 - Machine-to-Machine (M2M) communication
 - RFID and sensor networks
 - Connected devices and smart objects
 - Internet-connected things with intelligence
- **Driving Forces:**
 - Moore's Law enabling cheaper, more powerful processors
 - Advances in wireless communication
 - Cloud computing and data analytics
 - Falling costs of sensors and actuators

IoT and Digitization



Wearable Tech Devices

It's everywhere!



Healthcare

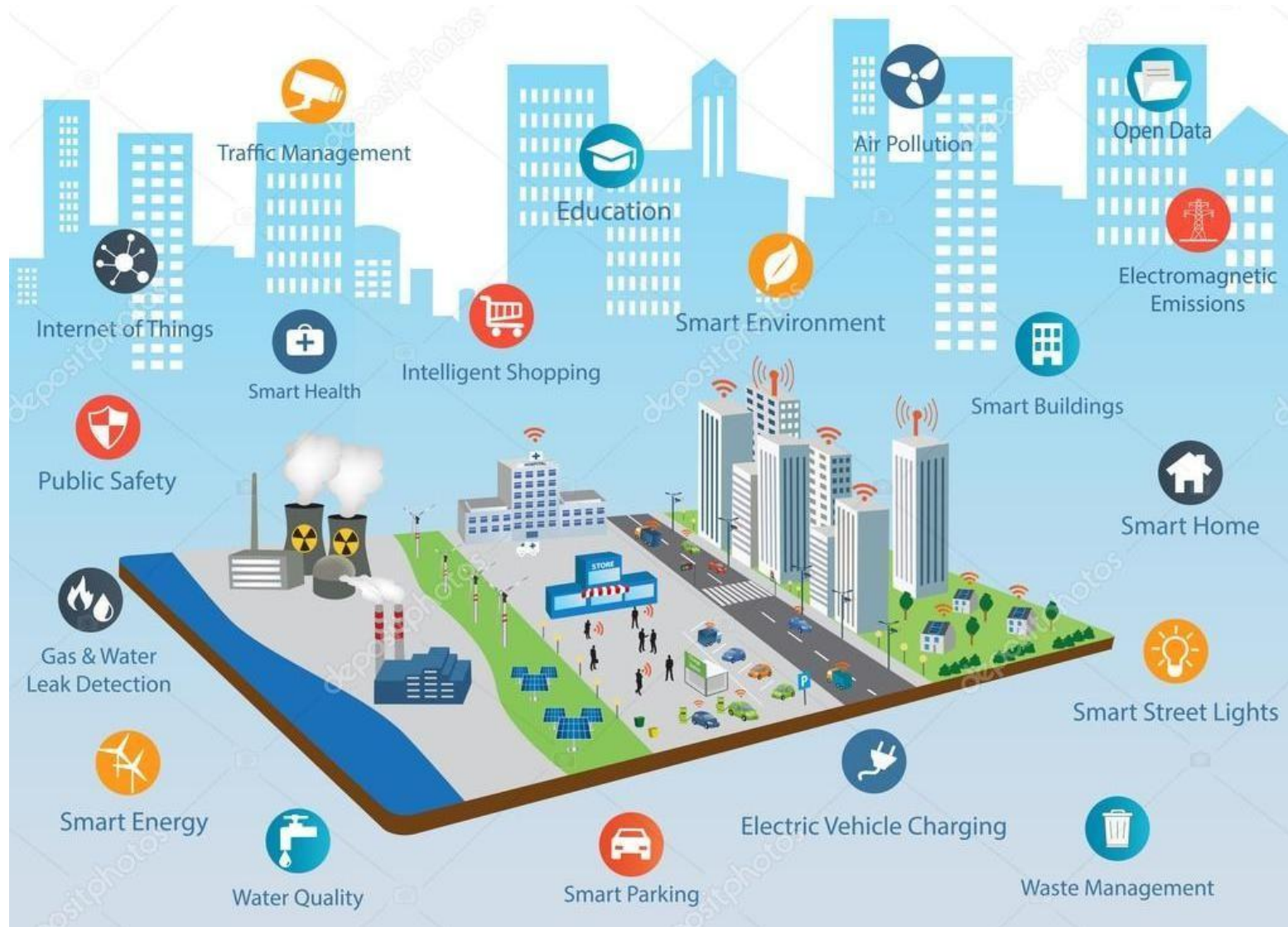


Industry Automation
and Monitoring



Smart Appliances

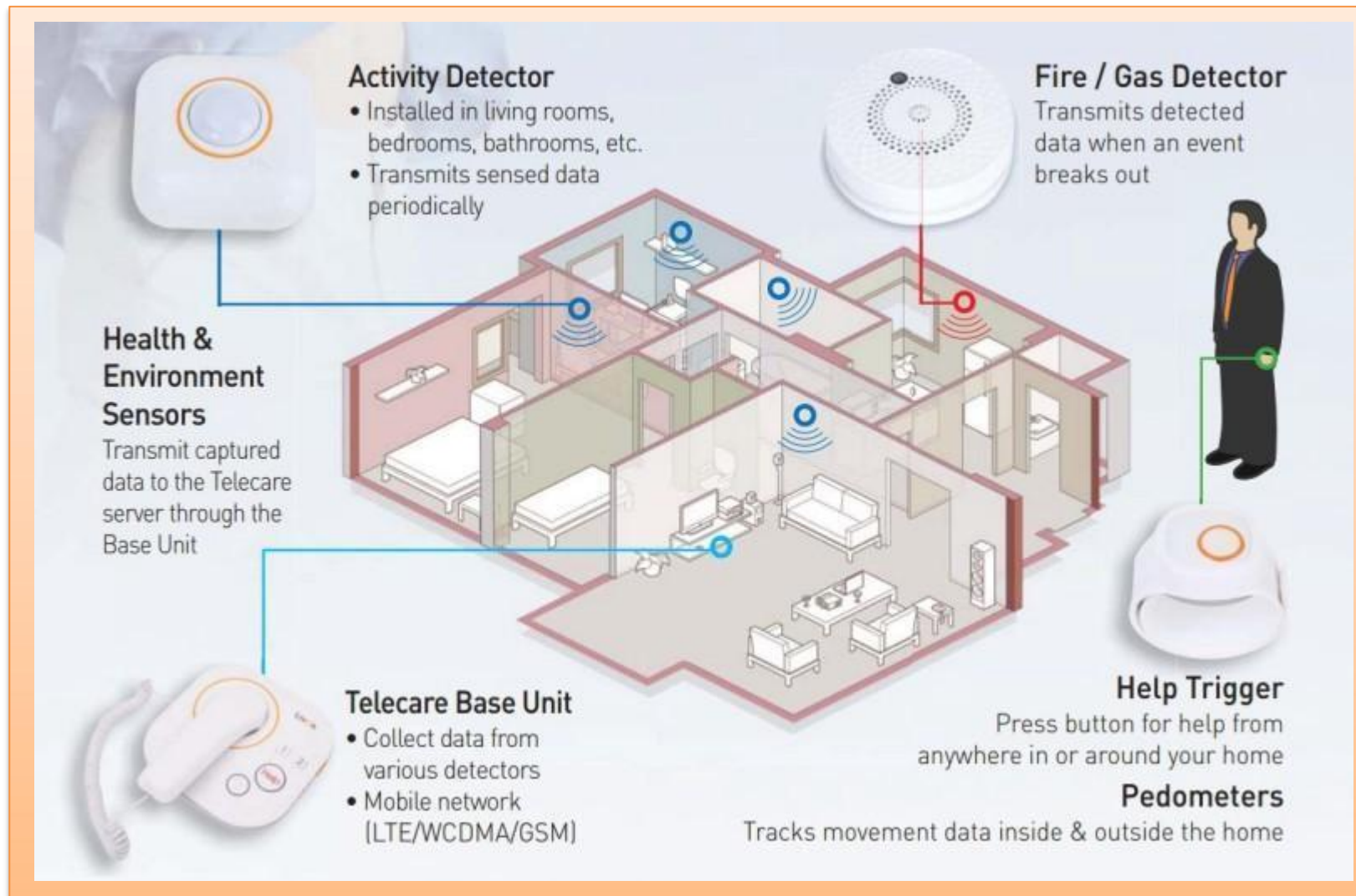
IoT and Digitization: Smart City



Source: <https://depositphotos.com/126025652/stock-illustration-smart-city-concept-and-internet.html>

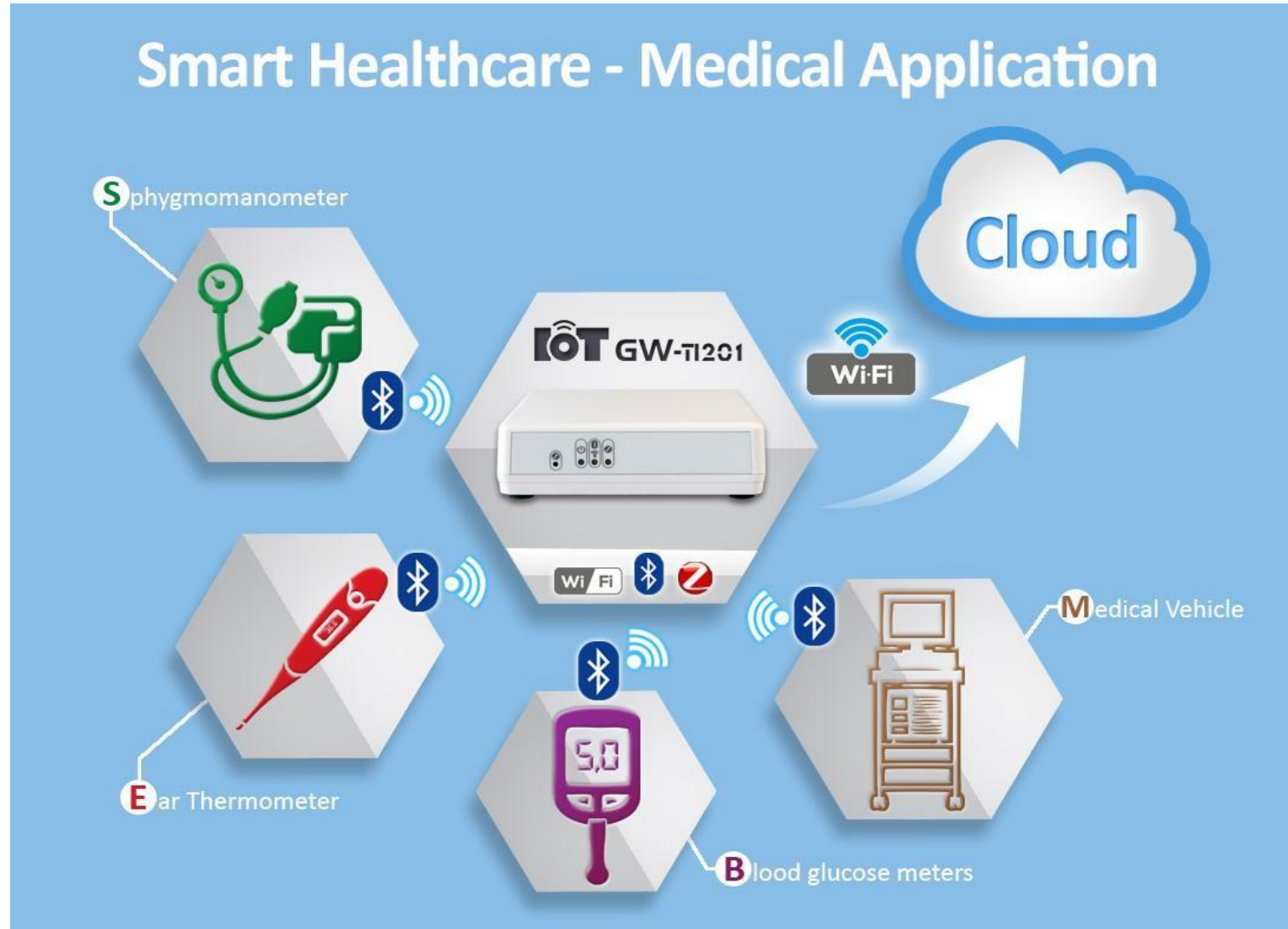
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IoT and Digitization: Smart Home



Source: <https://medium.com/@globalindnews/north-america-accounted-for-major-share-in-the-global-smart-home-healthcare-market-in-2015-cc9cc1874a5f>

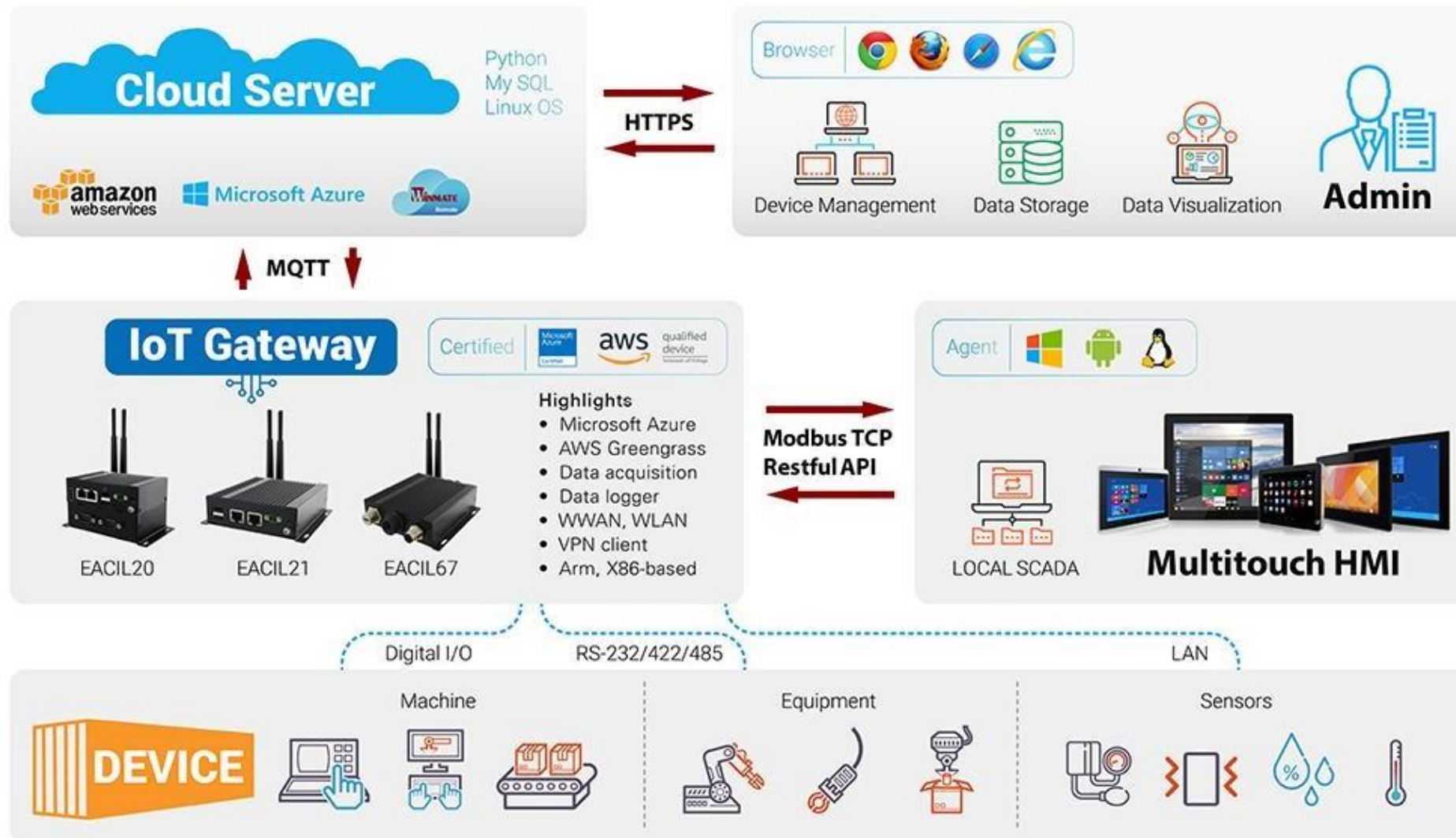
IoT and Digitization: Smart Healthcare



Source: <http://iot.fit-foxconn.com/>

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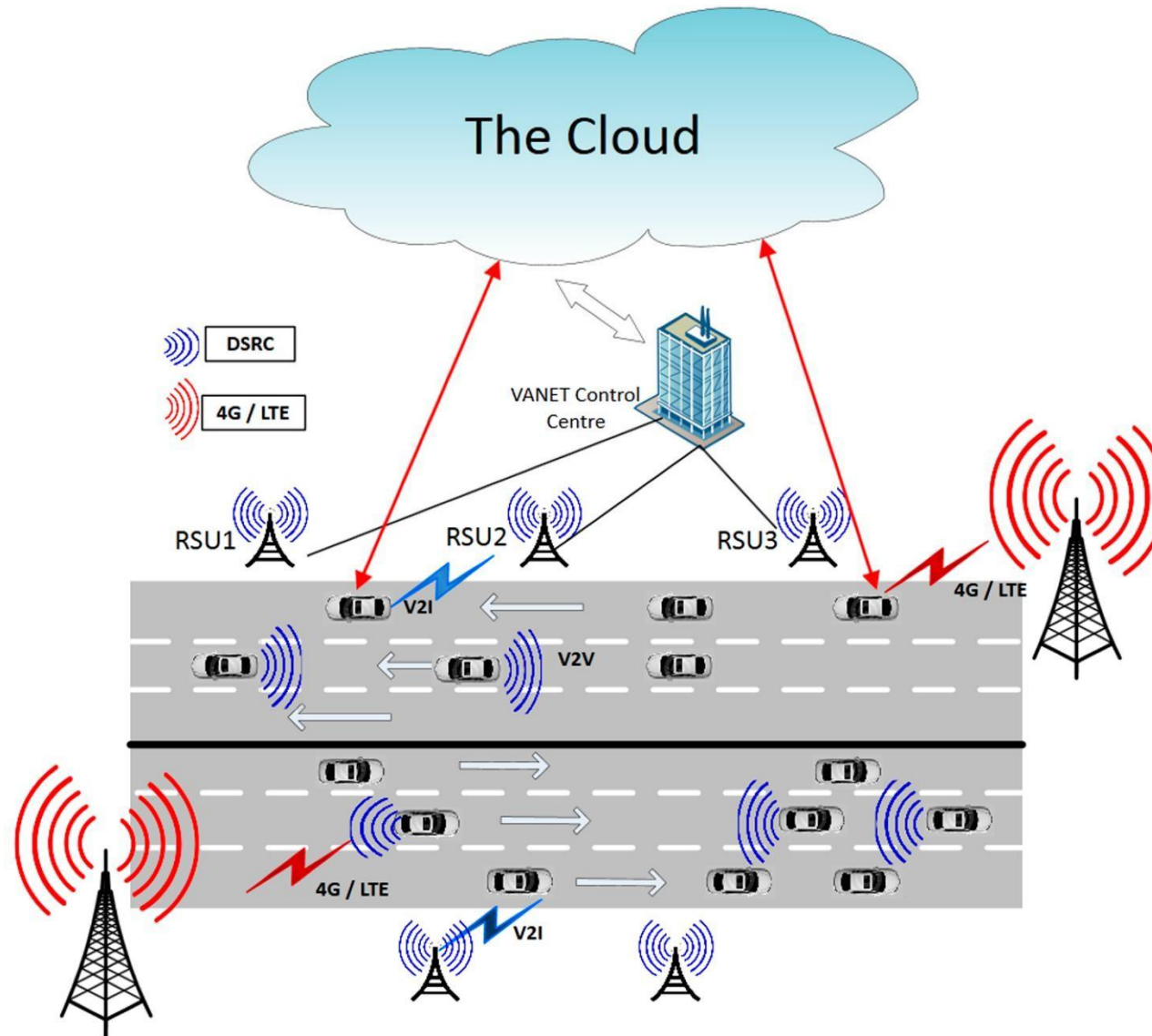
IoT and Digitization: Industrial IoT



Source: https://www.winmate.com/Solutions/Solutions_IoT.asp

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IoT and Digitization: Connected Cars



Source: Vehicular K. Ziadi, M. Rajarajan, "Internet: Security & Privacy Challenges and Opportunities", *Future Internet* 2015, 7(3), 257-275.

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IoT and Digitization: Google's Self-Driving Car



Source: <https://www.google.com/>
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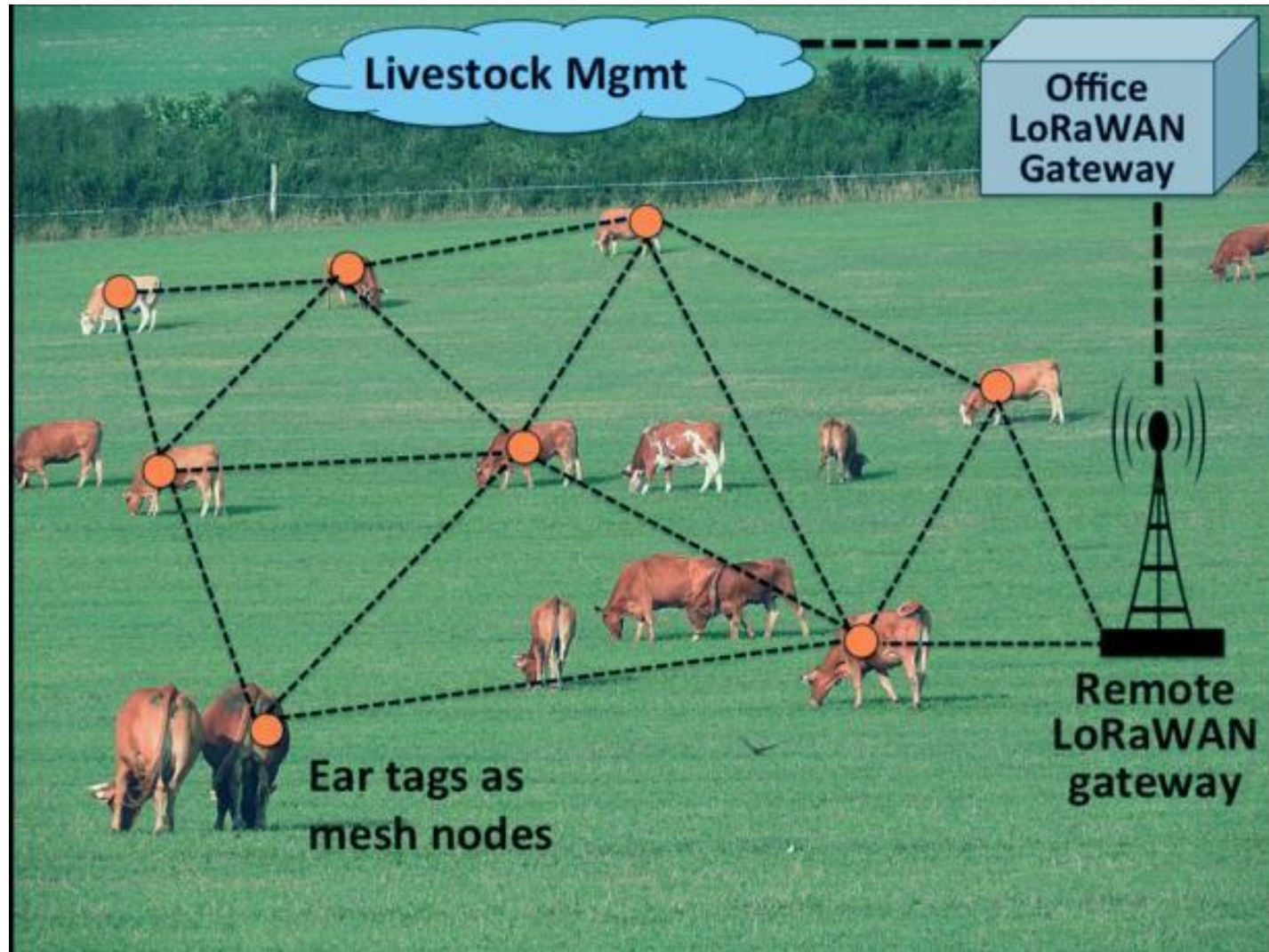
IoT and Digitization: Smart Agriculture



Source: <https://in.pinterest.com/pin/515380751093603767/?lp=true>

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IoT and Digitization: Livestock Management



Source: <https://data-flair.training/blogs/iot-applications-in-agriculture/>

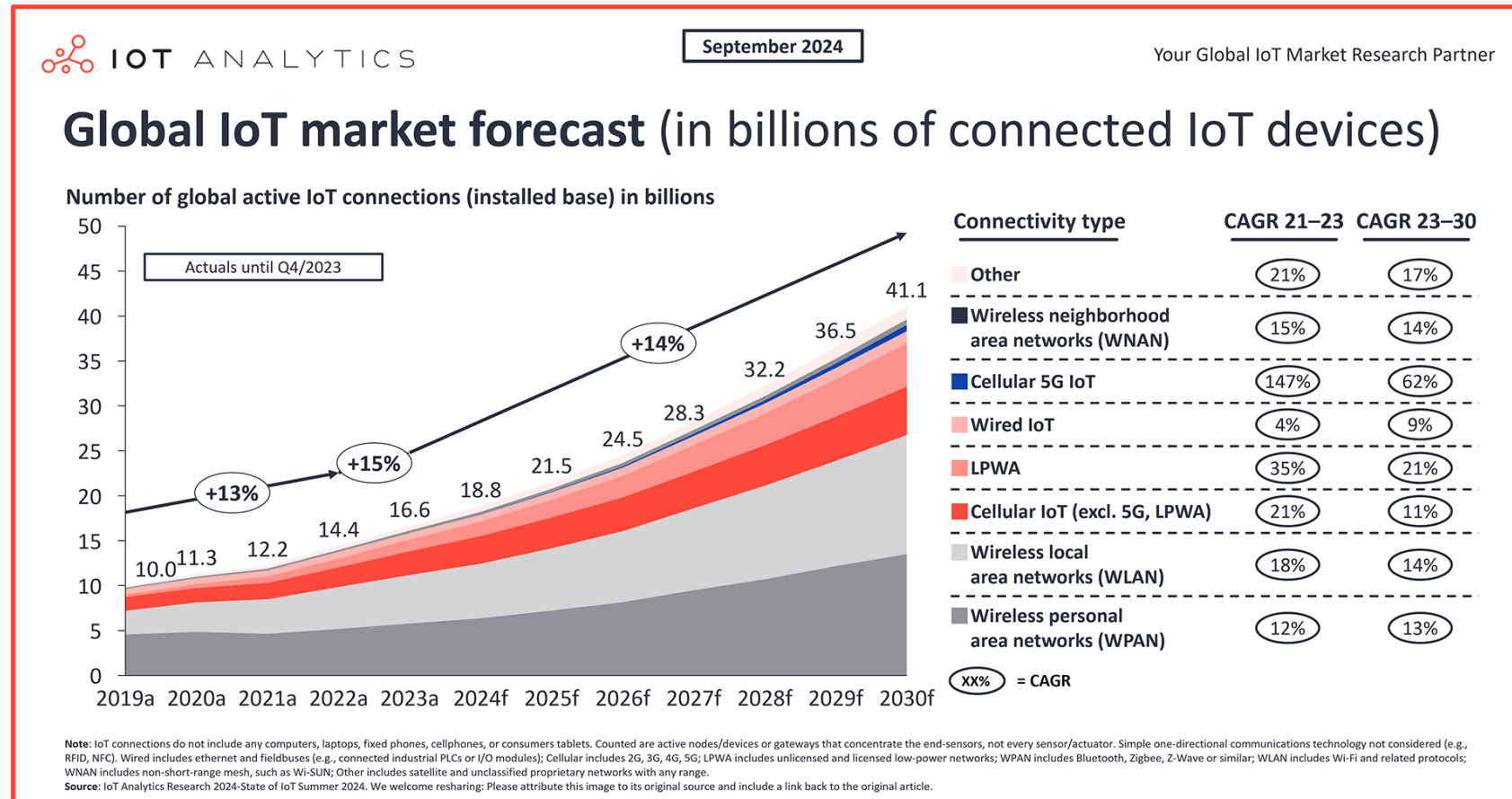
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IoT and Digitization (Summarized)

- **Digitization Impact:** Transforming physical assets into digitally connected entities
- **Connected Roadways:** Traffic monitoring, autonomous vehicles
- **Connected Factory:** Predictive maintenance, real-time monitoring
- **Smart Connected Buildings:** Energy efficiency, automated systems
- **Smart Creatures:** Wearable technology for livestock monitoring

IoT Impact and Growth



IoT Analytics' prediction

IoT devices excluding PC, Laptop, Smartphone, Tablet, and one directional technology like RFID, NFC

Image Source: <https://iot-analytics.com/number-connected-iot-devices/>

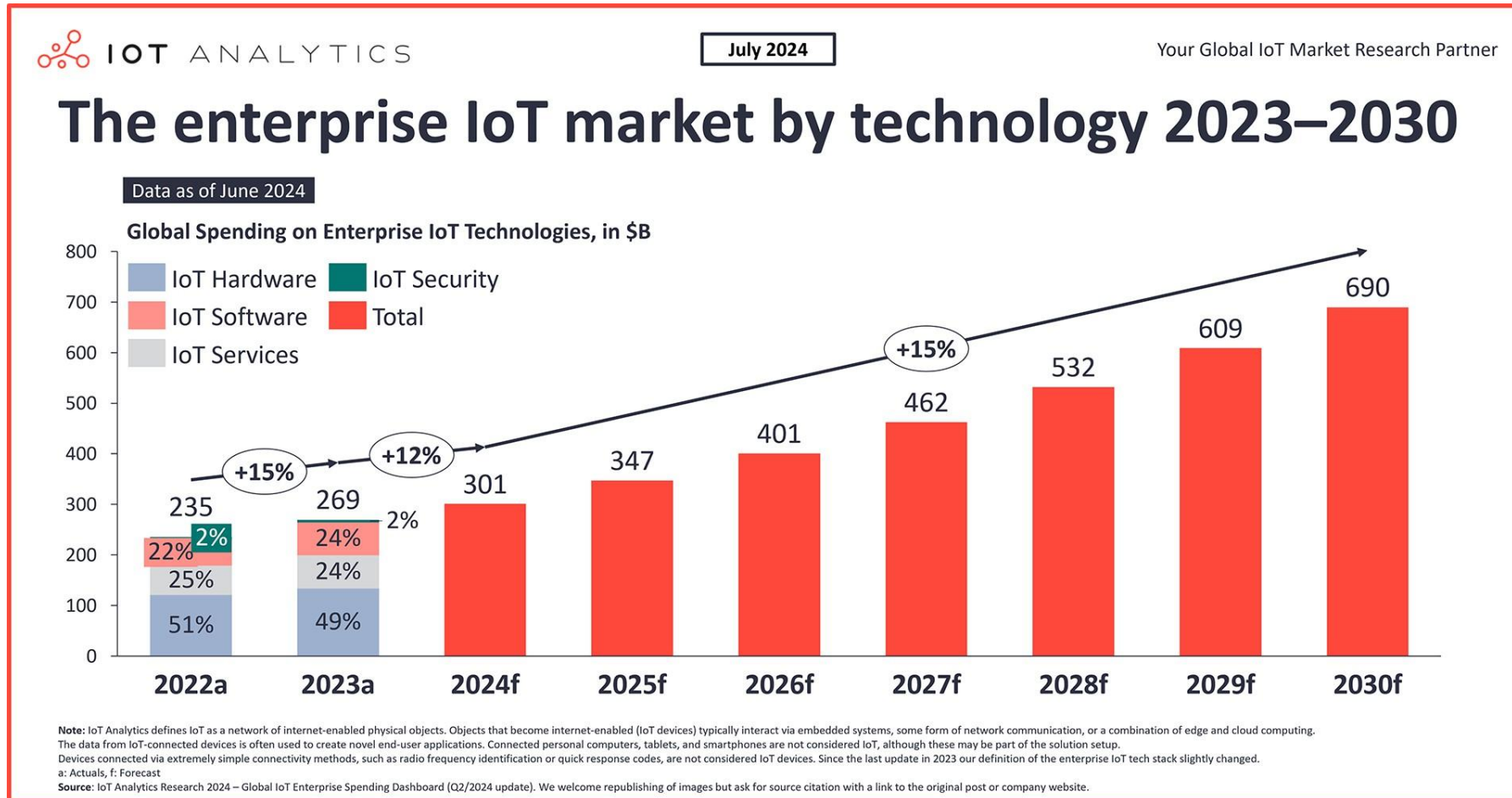
Wired: Eth, SCADA network (Modbus, Fieldbus, etc.)

Cellular: 2G, 3G, 4G, 5G; **LPWA:** LoRa, HaLow

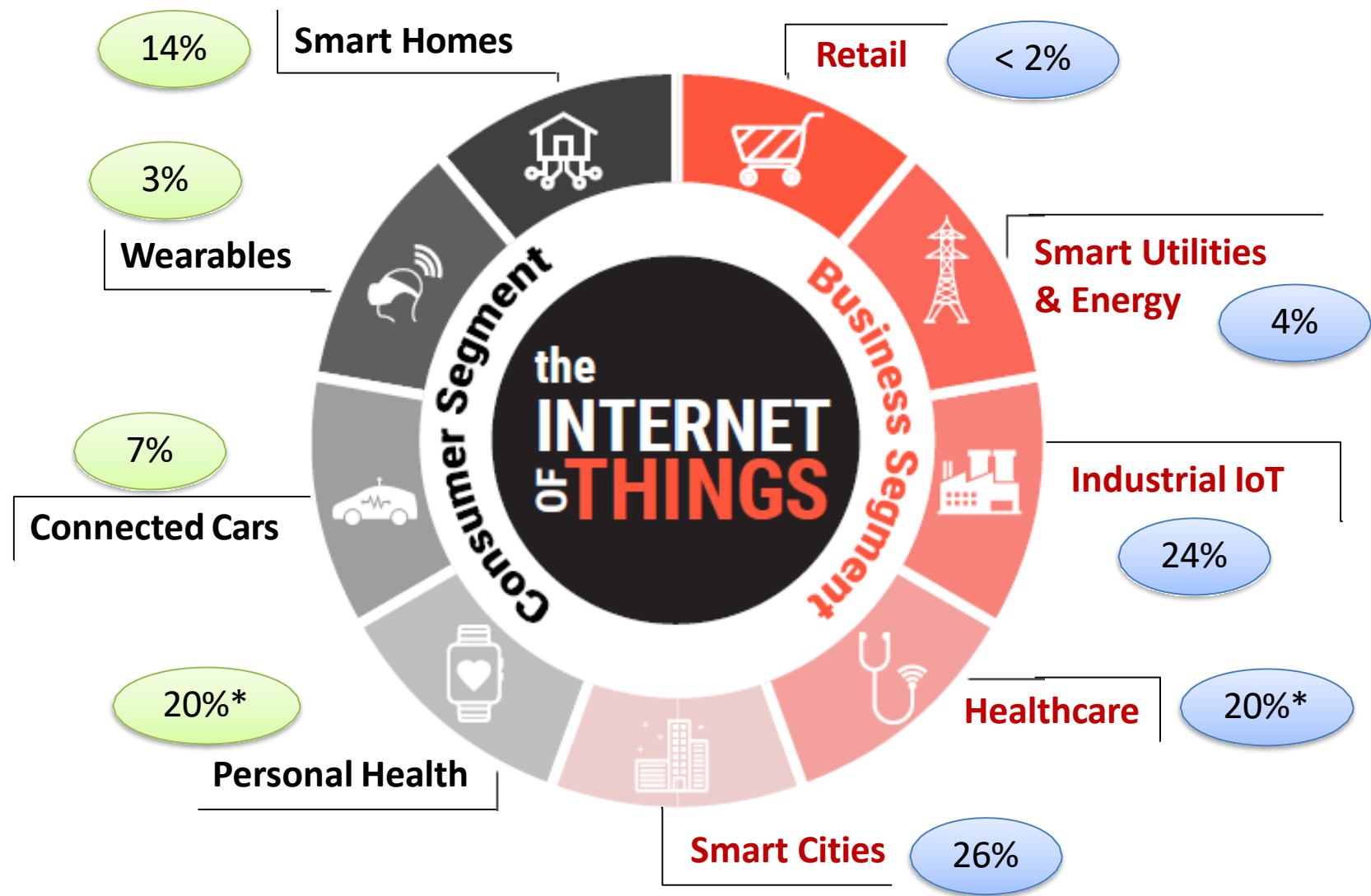
WPAN: BLE, ZigBee, Z-Wave, etc.; **WLAN:** Wi-Fi

WNAN: Non-short-range mesh e.g. Wi-SUN

IoT Impact and Growth...



IoT Impact and Growth...



Global Market Share of IoT
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Source: <https://growthenabler.com/flipbook/pdf/IOT%20Report.pdf>



IoT Impact and Growth...

- **Connected Devices Forecast:** Billions of devices internet-connected
- **Industry Impact Areas:**
 - Manufacturing
 - Energy and Utilities
 - Transportation
 - Healthcare
 - Smart Cities
 - Public Safety

IoT Impact and Growth...

➤ **Benefits of IoT**

➤ **Real-Time Monitoring**

- IoT sensors can monitor equipment and processes in real-time, providing valuable sensed data that can be used to streamline operations, reduce waste, and increase output.

➤ **Automation of Processes**

- Machines can assemble parts with more precision and speed, resulting in
 - fewer errors during assembly
- Robots can very rapidly detect faults that may not be detected by the human eye

➤ **Improved Efficiency or Productivity**

- IoT helps to enhance productivity by streamlining processes, automating tasks, and providing real-time data insights.

➤ **Predictive Maintenance**

- Continuous monitoring of systems and processes to identify key indicators of problems before they result in downtime or system failure

IoT Impact and Growth...

➤ **Benefits of IoT**

➤ Improved or New Insights

- IoT generates vast amounts of data that can be analyzed to identify trends, inefficiencies, and areas for improvement. Organizations can leverage these insights to enhance productivity and operational effectiveness.

➤ Cost Reduction

- When an organization can improve system uptime, automate processes, reduce the risk of failure and gain insights that support better decision making, and reduce resource usage, the result is efficiency and cost savings

➤ Optimized Work Environments

- IoT technology can help create more comfortable and efficient workspaces

➤ Adaptability

- The ability to adapt to new business requirements, customer needs, and changing conditions, or scale the deployment in response to business growth or customer requirements

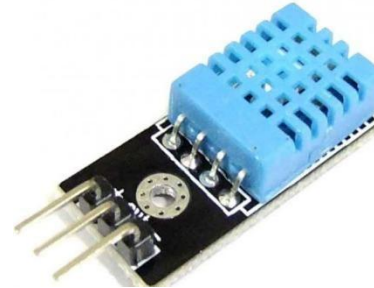
“Things” in IoT – Sensors



MQ135 - Air Quality Gas Sensor



Sound Detection Sensor



DHT11 - Temperature and Humidity Sensor



PIR Motion Detector Sensor



Pulse Sensor



LDR Light Sensor



Ultrasonic Distance Sensor



IR Sensor

“Things” in IoT – Actuators



4 Channel 5V Relay



Servo Motor



DC Motor



Solenoid valve



Linear Actuators



LED



LCD Display

Access Technologies in IoT

Communication Criteria

- Range
- Frequency Bands
- Power Consumption
- Topology
- Constrained Devices
- Constrained-Node Networks

IoT Access Technologies



IoT Enabling Technologies

- Enabling technologies are the foundational technological components and advancements that make the Internet of Things possible and practical.
- **IoT is driven by breakthroughs in:**
 - RFID (Radio Frequency Identification)
 - Smart Sensors
 - Communication Technologies
 - Internet Protocols

IoT Enabling Technologies...

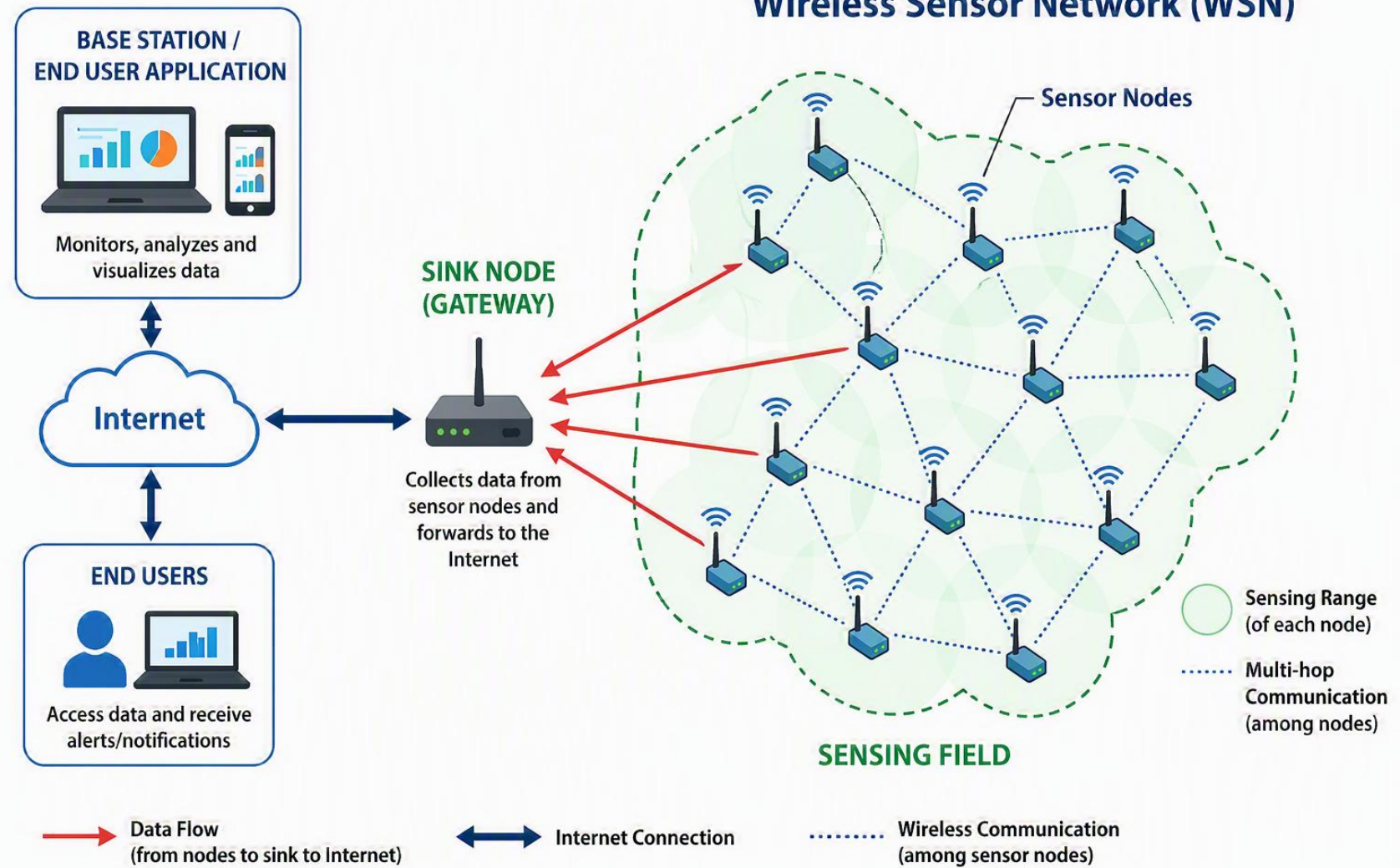
Overview of IoT Enabling Technologies

Technology Category	Primary Role
Wireless Sensor Networks (WSN)	Environmental sensing and data collection
RFID	Object identification and tracking
Embedded Systems	Local processing and device intelligence
Communication Protocols	Data exchange between devices and networks
Cloud Computing	Scalable storage and computational resources
Big Data Analytics	Extracting insights and enabling decisions

Enabling Technology 1 – Wireless Sensor Networks (WSN)

- A WSN is a network of spatially distributed autonomous sensors that monitor physical or environmental conditions (temperature, humidity, pressure, motion, etc.) and cooperatively pass their data through wireless links.
- **Key Functions:**
 - Distributed sensing and data acquisition
 - Self-organisation and fault tolerance
- **Role in IoT:**
 - Forms the **perception layer** of the IoT architecture
 - Provides raw data for upper-layer applications
- **Typical Applications:** Environmental monitoring, precision agriculture, industrial automation, smart cities.

Enabling Technology 1 – Wireless Sensor Networks (WSN)

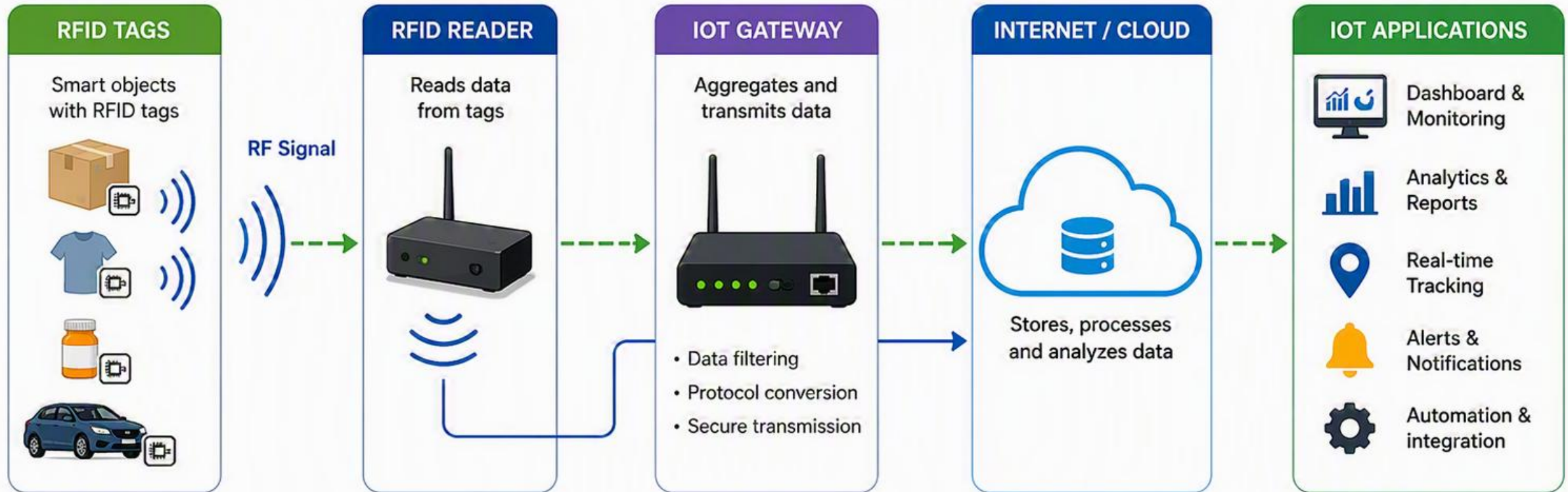


Enabling Technology 2 – Radio Frequency Identification (RFID)

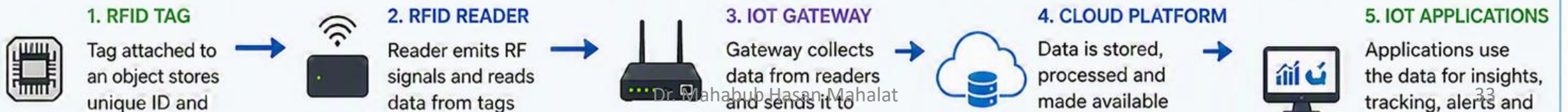
- RFID uses radio waves to automatically identify and track tags attached to objects.
- **Core Components:**
 - **RFID Tag** – stores identification data (passive, active, or semi-passive)
 - **RFID Reader** – reads from and writes to tags
- **Role in IoT:**
 - Enables unique identification and traceability of physical objects
 - **Sensor-enabled RFID tags** can also capture environmental data
 - Passive tags enable battery-free sensing
- **Typical Applications:** Supply chain management, asset tracking, access control, smart logistics.

RFID in IoT

Connecting Physical Objects to the Internet for Smart Identification and Tracking



HOW IT WORKS



Enabling Technology 3 – Embedded Systems

- An embedded system is a dedicated computer system designed to perform a specific function within a larger mechanical or electrical system.
- **Core Components:**
 - **Processor** (microcontroller or microprocessor)
 - **Memory** (RAM, ROM, Flash)
 - **I/O Interfaces** (sensors, actuators, communication ports)
- **Role in IoT:**
 - Acts as the “brain” of each IoT device
 - Modern Systems-on-Chip (SoC) integrate processing, connectivity, and even AI acceleration
- **Common Platforms:** Arduino, Raspberry Pi, ESP8266/ESP32, ARM Cortex-M based MCUs.

EMBEDDED SYSTEMS

Embedded systems are dedicated computer systems designed to perform specific tasks within larger systems.

INDUSTRIAL ROBOTS



Used for automation and precise control in manufacturing.

GPS RECEIVERS



Provides location and navigation information.

DIGITAL CAMERAS



Captures images and processes data.

DVD PLAYERS



Reads discs and handles playback operations.

WIRELESS ROUTERS



Manages network traffic and enables connectivity.

SET TOP BOXES



Receives and decodes TV signals for display.

GAMING CONSOLES



Processes graphics, input and provides gaming experience.

PHOTOCOPIERS



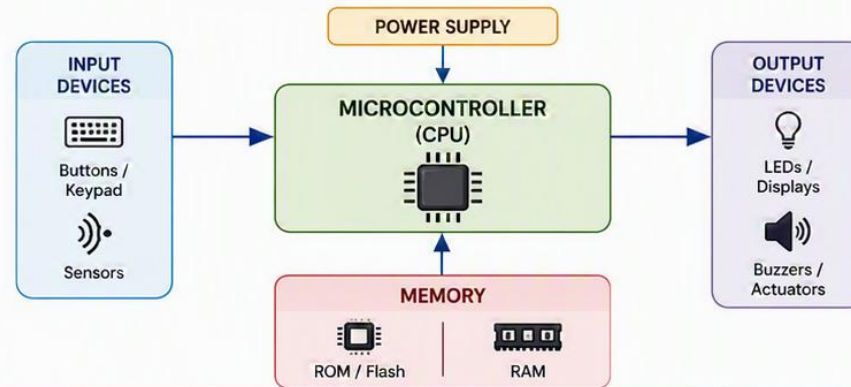
Scans, copies and prints documents.

MICROWAVE OVENS



Heats food and manages time, temperature and safety.

TYPICAL EMBEDDED SYSTEM ARCHITECTURE

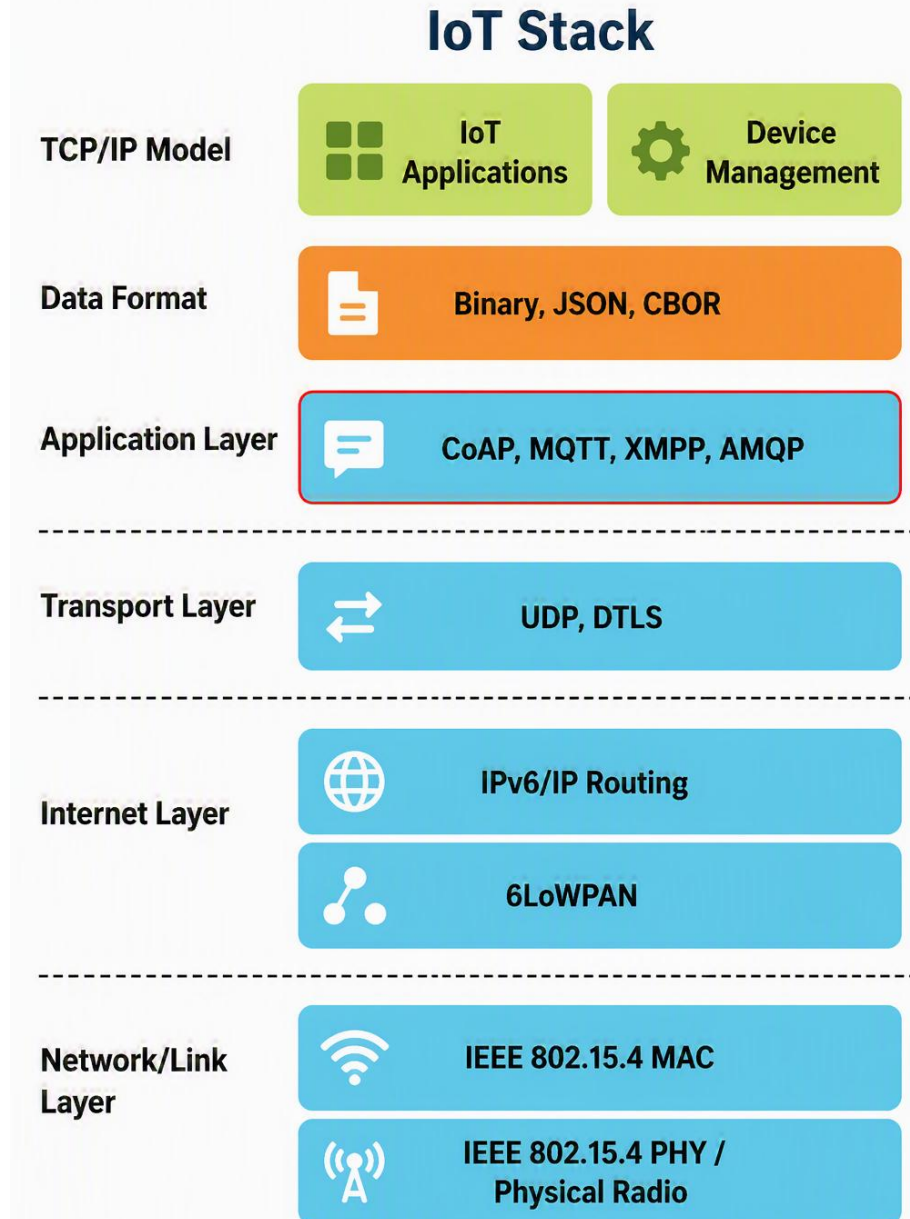


Enabling Technology 4 – Communication Protocols

- Communication protocols define the rules and formats for data exchange between IoT devices and networks.
- **Protocol Stack (Layered View):**
- **Key IoT-specific Protocols:**
 - **MQTT** – lightweight publish/subscribe messaging
 - **CoAP** – RESTful protocol for constrained devices
 - **6LoWPAN** – IPv6 over low-power wireless personal area networks

Layer	Example Protocols
Link Layer	IEEE 802.3 (Ethernet), 802.11 (Wi-Fi), 802.15.4 (ZigBee), Bluetooth LE
Network Layer	IPv4, IPv6, 6LoWPAN
Transport Layer	TCP, UDP
Application Layer	HTTP, MQTT, CoAP, AMQP

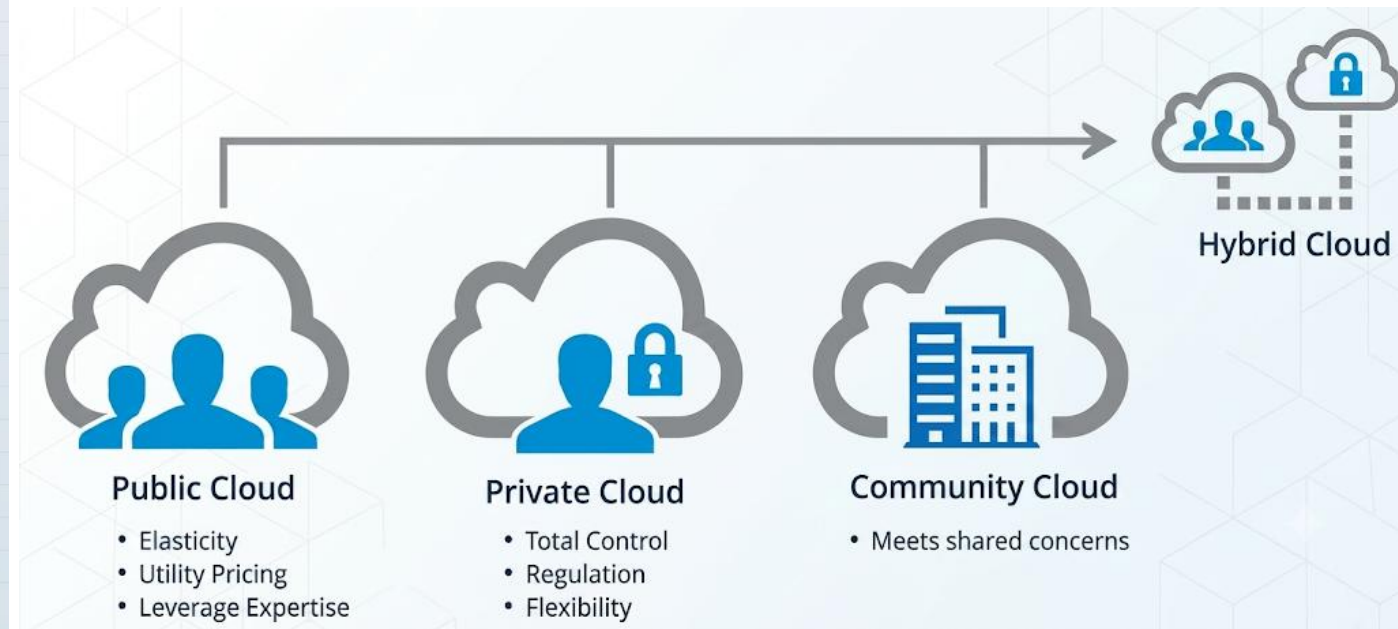
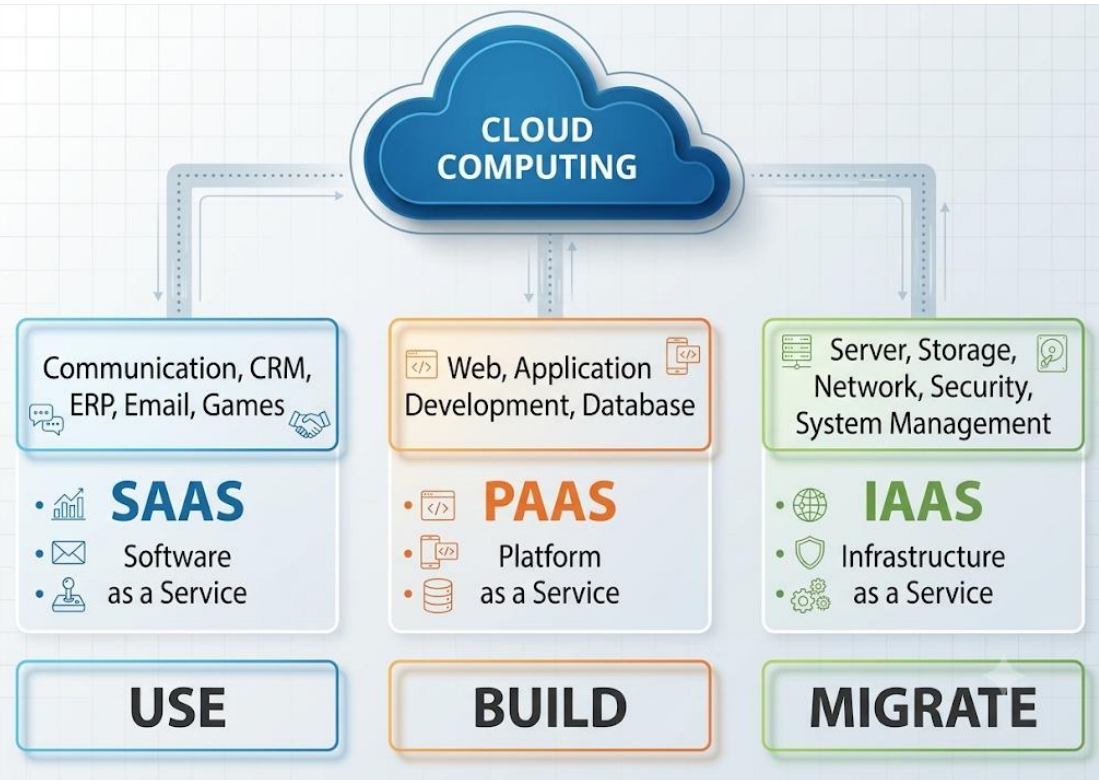
Enabling Technology 4 – Communication Protocols



Enabling Technology 5 – Cloud Computing

- Cloud computing delivers computing services (servers, storage, databases, networking, software) over the internet (“the cloud”).
- **Core Resources:** Compute, Networking, Storage.
- **Service Models:**
 - **IaaS** – Infrastructure as a Service
 - **PaaS** – Platform as a Service
 - **SaaS** – Software as a Service
- **Deployment Models:**
 - Public, Private, Community, and Hybrid clouds
- **Role in IoT:**
 - Provides virtually unlimited and scalable storage and processing power
 - Enables on-demand access to resources, reducing upfront capital expenditure
 - Supports device management and application hosting

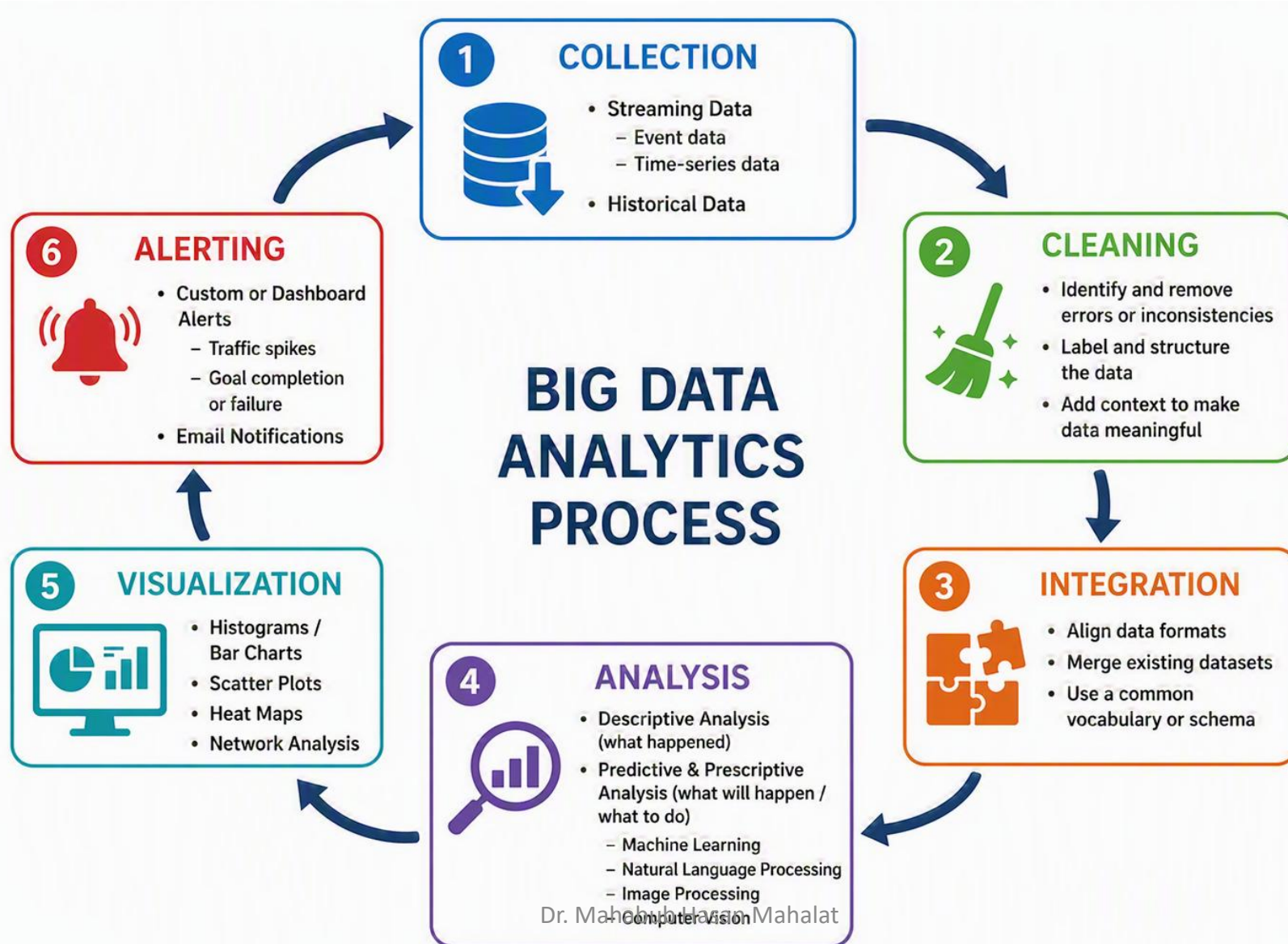
Enabling Technology 5 – Cloud Computing



Enabling Technology 6 – Big Data Analytics

- Big Data analytics involves collecting, processing, and analysing large and diverse datasets (volume, velocity, variety) to uncover patterns, correlations, and insights.
- **Role in IoT:**
 - Transforms massive sensor data streams into actionable intelligence
 - Enables real-time monitoring and predictive maintenance
- **Key Concepts:**
 - Structured vs. unstructured data
 - Data in motion (streaming) vs. data at rest (batch)
- **Technologies & Tools:**
 - NoSQL databases (MongoDB, Cassandra)
 - Hadoop ecosystem, Apache Spark, Apache Kafka
 - Machine learning frameworks (TensorFlow, Scikit-learn)
- **Value Proposition:** Turning raw data into decisions – the ultimate goal of IoT.

Enabling Technology 6 – Big Data Analytics



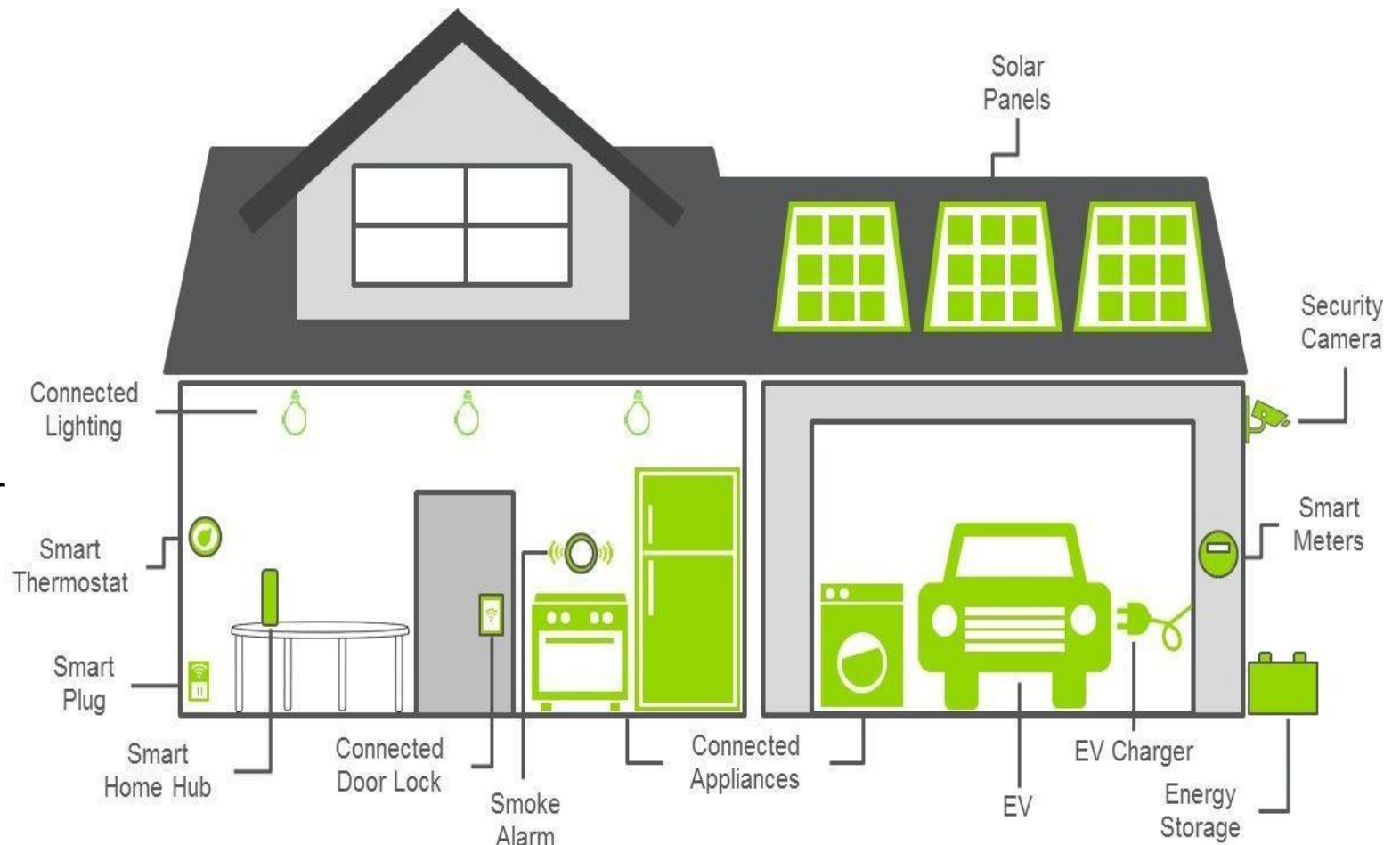
IoT Network Architecture and Design

Architectural Plan

- In present days, networks run the modern business
- So, it should never be built without careful planning
- Architecture is how you design your application or solution.
- **Essence of IoT architecture:**
 - how the data is
 - transported,
 - collected,
 - analyzed, and
 - ultimately acted upon.

Design Challenges of IoT Architecture:

- Scale
- Security
- Constrained devices
- Massive data
- Data analysis
- Support to legacy devices



Smart Home

Emergence of Standard IoT Architecture

- The basic 3-layer or commonly used 4-layer architectures of IoT did not address many issues.
- IoT did not have a standard defined architecture of working, which is strictly followed universally.
- Existing Issues:
 - Highly fragmented marked with limited vendor-specific applications
 - Each application vertical contains its own technologies without interoperability
 - Incompatibility for seamless integration between heterogeneous applications and devices
 - Did not consider security aspect in design
- In the past several years, few architectural standards and frameworks have emerged
- Two best-known architectures:
 - **oneM2M architecture**
 - **IoT World Forum architecture**



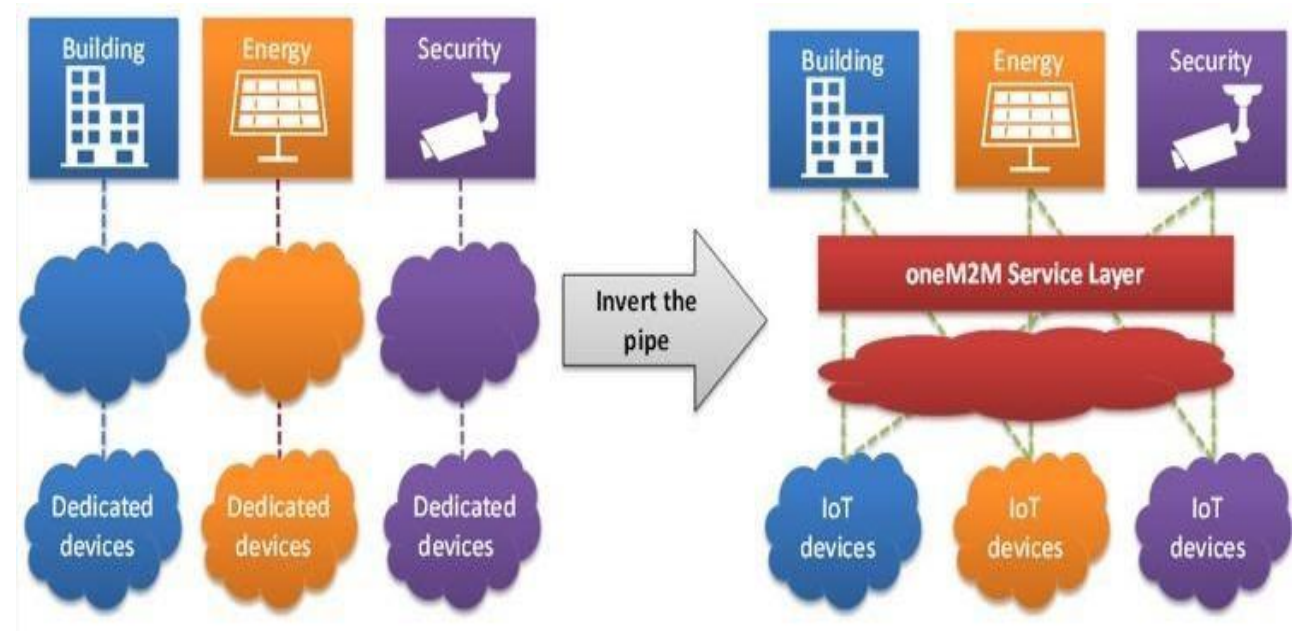
oneM2M Architecture

- Goal of **oneM2M** architecture: to create a common services layer, which can be readily embedded in **field devices** to allow communication with application servers.
- **Global initiative (2012)** – Launched by ETSI + 13 partners to standardize M2M/IoT globally
- **Common services layer** – Embedded in devices to enable uniform communication with application servers
- **Horizontal platform** – Solves device/software/network heterogeneity across all IoT stack levels
- **RESTful APIs** – Bridge incompatible systems (e.g., LoRaWAN ↔ BACnet) for seamless end-to-end interoperability

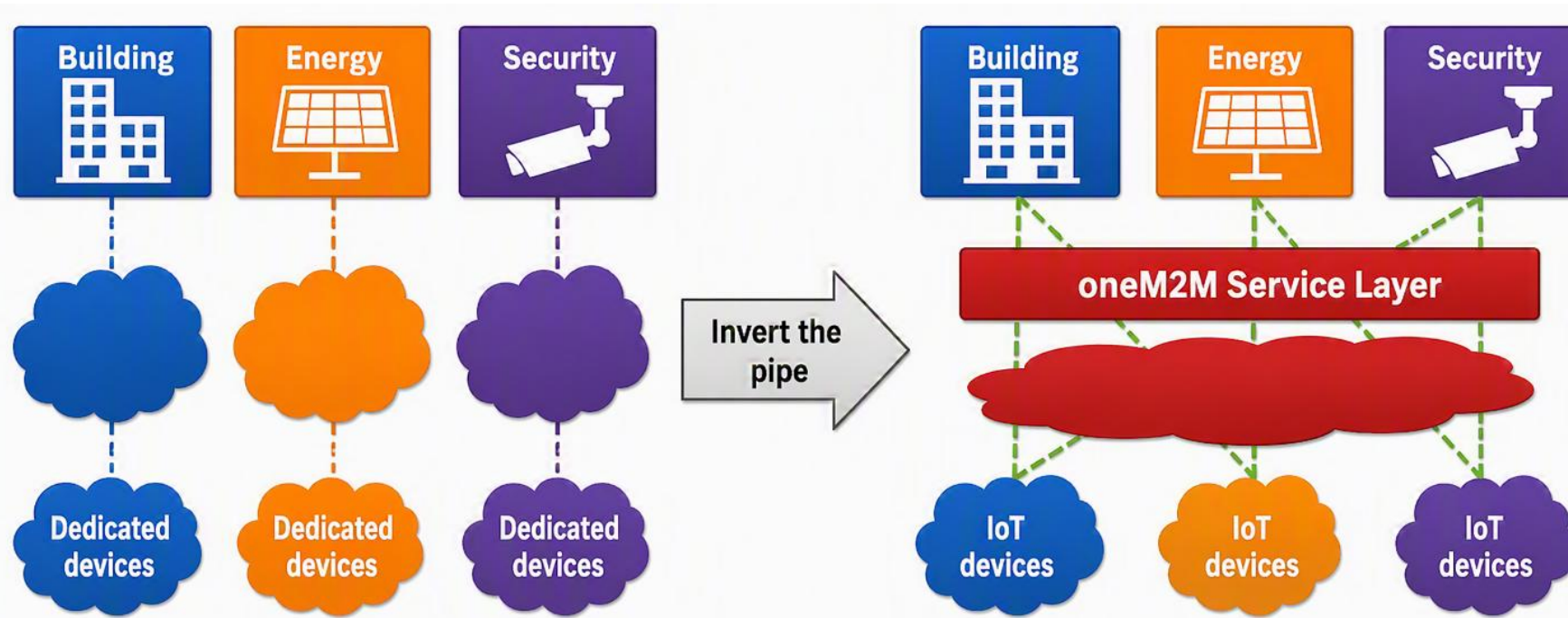
- Using the smart building use case, a **security application** can detect when nobody is in the building.
- It could then **trigger lights** to be switched off and for the **HVAC system** to operate on a reduced setting.

Challenges in IoT Architecture:

- Heterogeneity of devices,
- Heterogeneity of software,
- Heterogeneity of access methods



oneM2M Architecture...



Without oneM2M

- Highly fragmented market with limited vendor-specific applications
- Reinventing the wheel: Same services developed again and again
- Each silo contains its own technologies without interoperability

With oneM2M

- End-to-end platform: common service capabilities layer
- Interoperability at the level of data and control exchanges via uniform APIs
- Seamless interaction between heterogeneous applications and devices

oneM2M Architecture...

➤ **oneM2M Provides Three-Layer Functional Architecture**

- **Application Layer** – Industry-specific vertical applications (smart grid, e-health, connected vehicles, etc.) with northbound APIs for BI system integration
- **Services Layer (Horizontal Framework)** – Common services layer with APIs and middleware; bridges diverse devices to cloud/data center application servers; supports multiple backhaul technologies (cellular, MPLS, VPN)
- **Network Layer** – Communication domain covering IoT devices, gateways, and connectivity technologies (IEEE 802.15.4, 802.11ah, PLC, etc.); gateway acts as demarcation point between device and core network

1.

oneM2M – Network Layer

Communication domain that connects IoT devices and gateways using diverse access and backhaul technologies.

Key Functions

- Connectivity for IoT devices
- Data transport to gateway
- Protocol adaptation
- Demarcation between device domain and core network

Components

-  IoT Devices (sensors, actuators)
-  IoT Gateway
-  Access Networks
-  Backhaul / Core Networks

IoT Devices



Temperature Sensor



Humidity Sensor



Smart Meter



Camera



Actuator

IEEE 802.15.4 | 802.11ah | PLC
Bluetooth LE | LoRaWAN | Ethernet | etc.

IoT Gateway



Backhaul / Core Network



Cellular (4G/5G)



MPLS



VPN / Internet

Others ...

Device Domain
(Field Area Network)

Demarcation Point
(Gateway)

Core Network Domain
(Wide Area Network)

2.

oneM2M – Services Layer (Horizontal Framework)

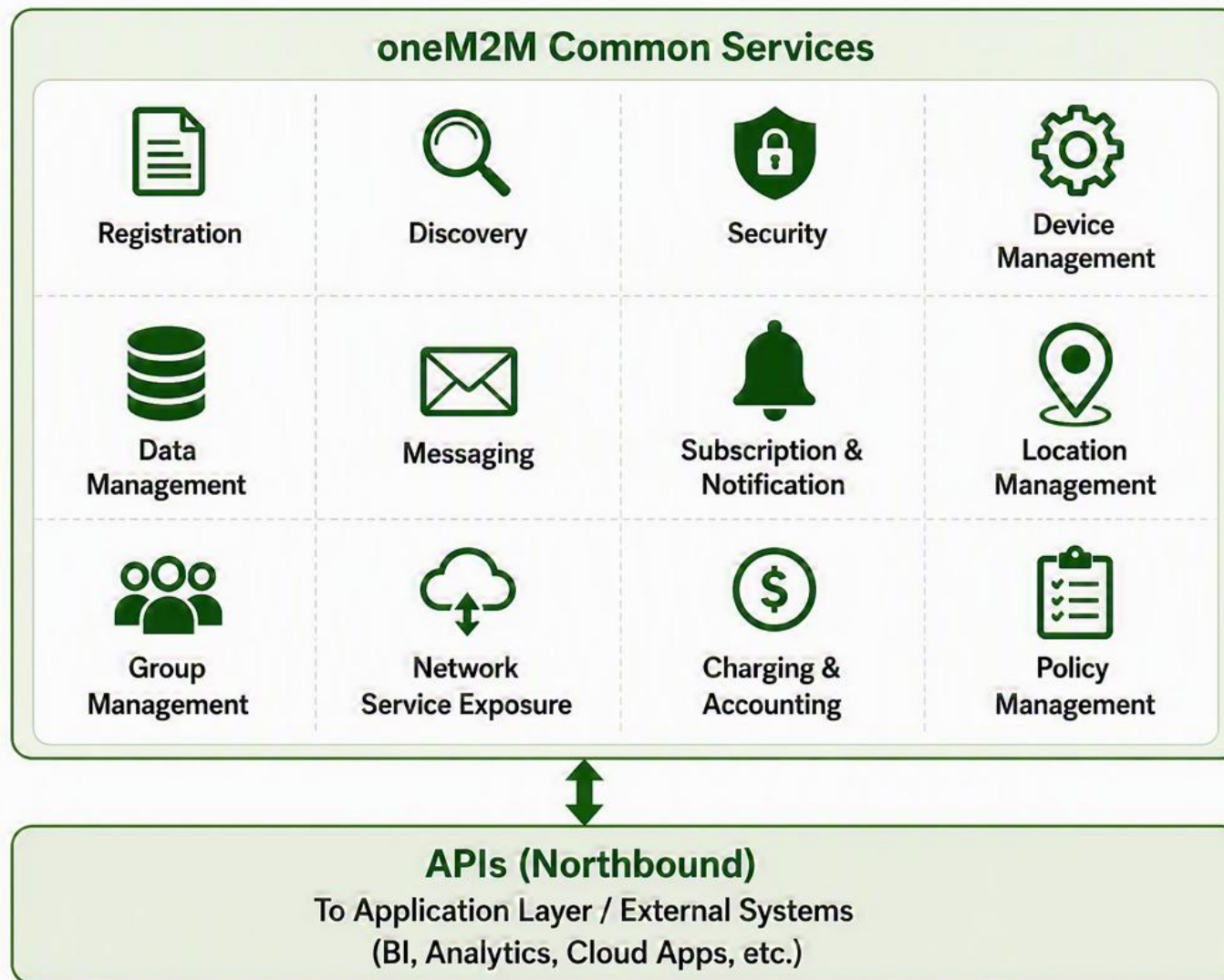
Common services layer with APIs and middleware that provides reusable functionalities and bridges diverse devices to application servers.

Key Functions

- Device management
- Security (authentication, authorization, integrity)
- Data management & storage
- Communication management
- Event management
- Subscription & notification
- Resource management
- Discovery & registration

Highlights

- Technology and industry agnostic
- Bridges diverse devices to application servers
- Exposes APIs (Northbound)
- Supports multiple backhaul technologies



3

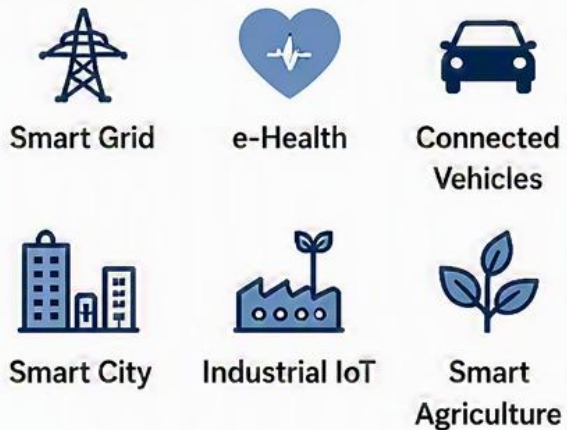
oneM2M – Application Layer

Industry-specific vertical applications that consume services exposed by the services layer and provide value to end users.

Key Functions

- Implement domain-specific use cases
- Consume services via northbound APIs
- Provide dashboards, analytics, alerts, and control
- Integrate with enterprise / BI systems

Examples



Application Capabilities

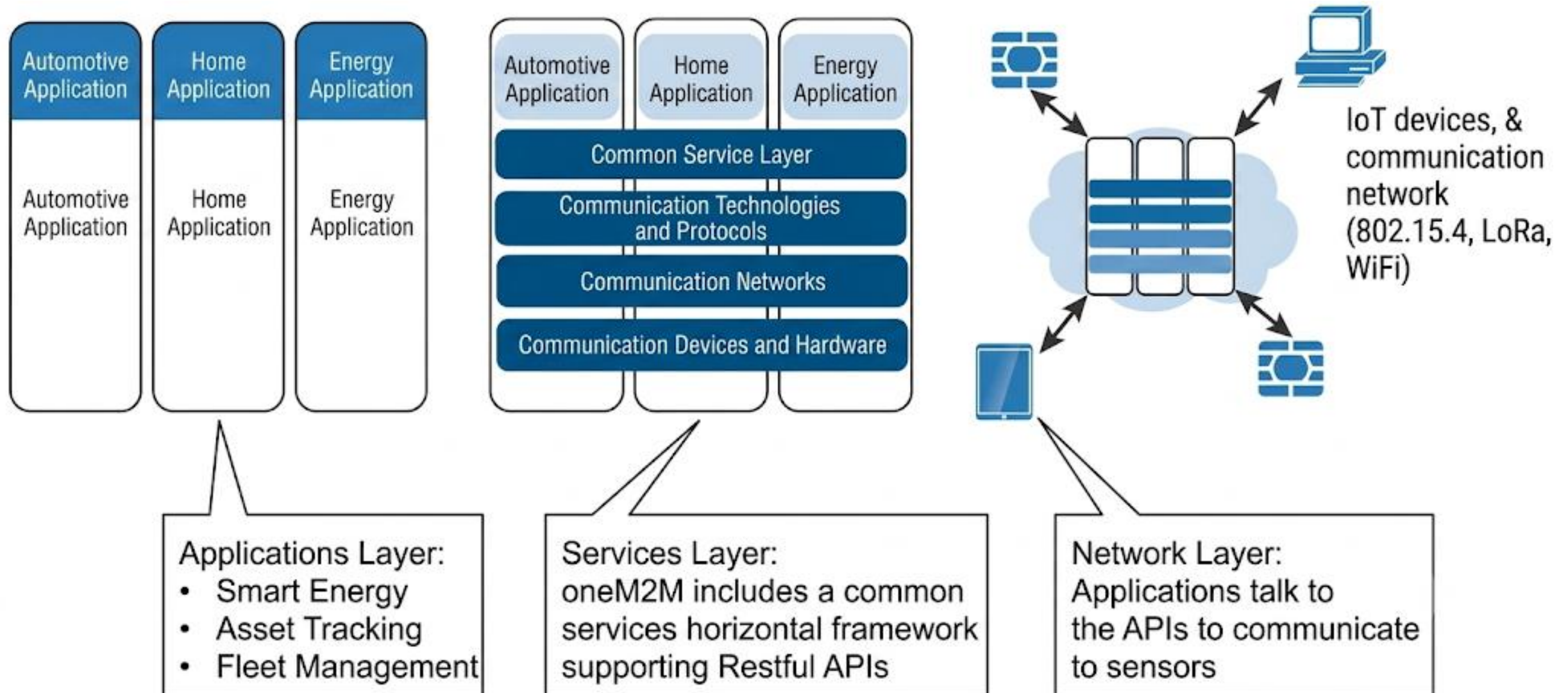


Northbound Integrations



Delivers actionable insights and automated actions for better decision making and operational efficiency.

oneM2M Architecture Summarized



IoT World Forum (IoTWF) Standardized Architecture

➤ Origin and Purpose:

- Published in **2014** by the IoTWF architectural committee
- Led by a consortium including **Cisco, IBM, Rockwell Automation,** and others
- Provides a **clean, simplified, seven-layer reference model** for IoT systems

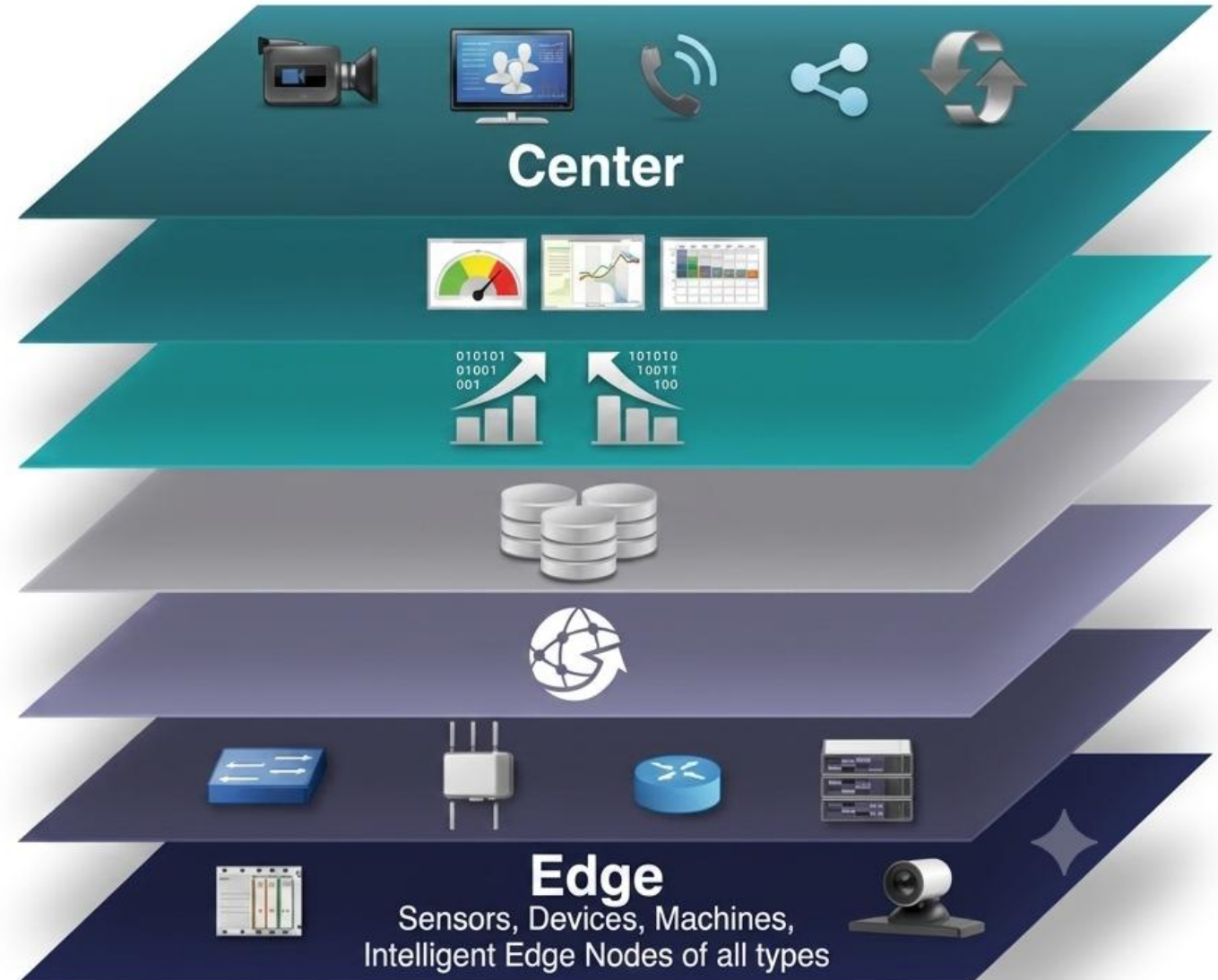
➤ Key Objectives:

- Decompose the IoT problem into manageable layers
- Ensure **interoperability** and **scalability** across diverse IoT deployments
- Define clear functions for each layer from sensors to applications
- **Security** encompasses the entire model

Architecture Overview – The Seven Layers

Levels

- 7 Collaboration & Processes**
(Involving People & Business Processes)
- 6 Application**
(Reporting, Analytics, Control)
- 5 Data Abstraction**
(Aggregation & Access)
- 4 Data Accumulation**
(Storage)
- 3 Edge Computing**
(Data Element Analysis & Transformation)
- 2 Connectivity**
(Communication & Processing Units)
- 1 Physical Devices & Controllers**
(The “Things” in IoT)



Layer 1 – Physical Devices and Controllers

➤ This layer is home to the "**things**" in the Internet of Things – the endpoint devices and sensors that generate data and can be controlled over the network

➤ **Key Characteristics:**

Aspect	Description
Scope	Wide range of endpoint devices that send and receive information
Size	From tiny silicon chips to large factory machines
Function	Generate data and respond to control commands
Diversity	Dozens/hundreds of manufacturers; no rules about form factor or origin

Layer 1 – Physical Devices and Controllers...

➤ Primary Responsibilities:

1.Data Collection – Sensing physical parameters (temperature, speed, health information, etc.)

2.Event-Based Processing – IoT devices and controllers perform event-based data aggregation and processing

3.Control Capability – Devices can be controlled remotely over the network

4.Data Transmission – Process information and transmit to controllers; send/receive data through global network infrastructure

➤ Level 1 devices do not typically include computing capabilities themselves – they rely on upper layers for processing.

➤ Examples:

➤ Temperature sensors

➤ Motion detectors

➤ Industrial machines

➤ Smart meters

➤ Wearable devices

Layer 2 – Connectivity

- This layer functions as the **transportation system** for IoT data, bridging the gap between edge devices and the cloud/data center.
- **Primary Function: Reliable and timely information transmission**
- **Three Types of Transmission:**
 - Between devices (Level 1) and the network
 - Across networks (east-west)
 - Between the network (Level 2) and low-level information processing at Level 3

2 Connectivity (Communication and Processing Units)

Layer 2 Functions:

- Communications Between Layer 1 Devices
- Reliable Delivery of Information Across the Network
- Switching and Routing
- Translation Between Protocols
- Network Level Security



SWITCH
(Layer 2 Device)



BRIDGE
(Layer 2 Device)



ROUTER
(Layer 3 Device)



MULTILAYER SWITCH
(Layer 2 and Layer 3 Device)

Layer 2 – Connectivity...

➤ Supported Communication Technologies:

Category	Technologies
Wireless	Wi-Fi (802.11), ZigBee (802.15.4), Bluetooth LE, LoRaWAN, NB-IoT, LTE-M
Wired	Ethernet (802.3), Power Line Communications (IEEE 1901)
Cellular	2G/3G/4G/5G

➤ Key Considerations:

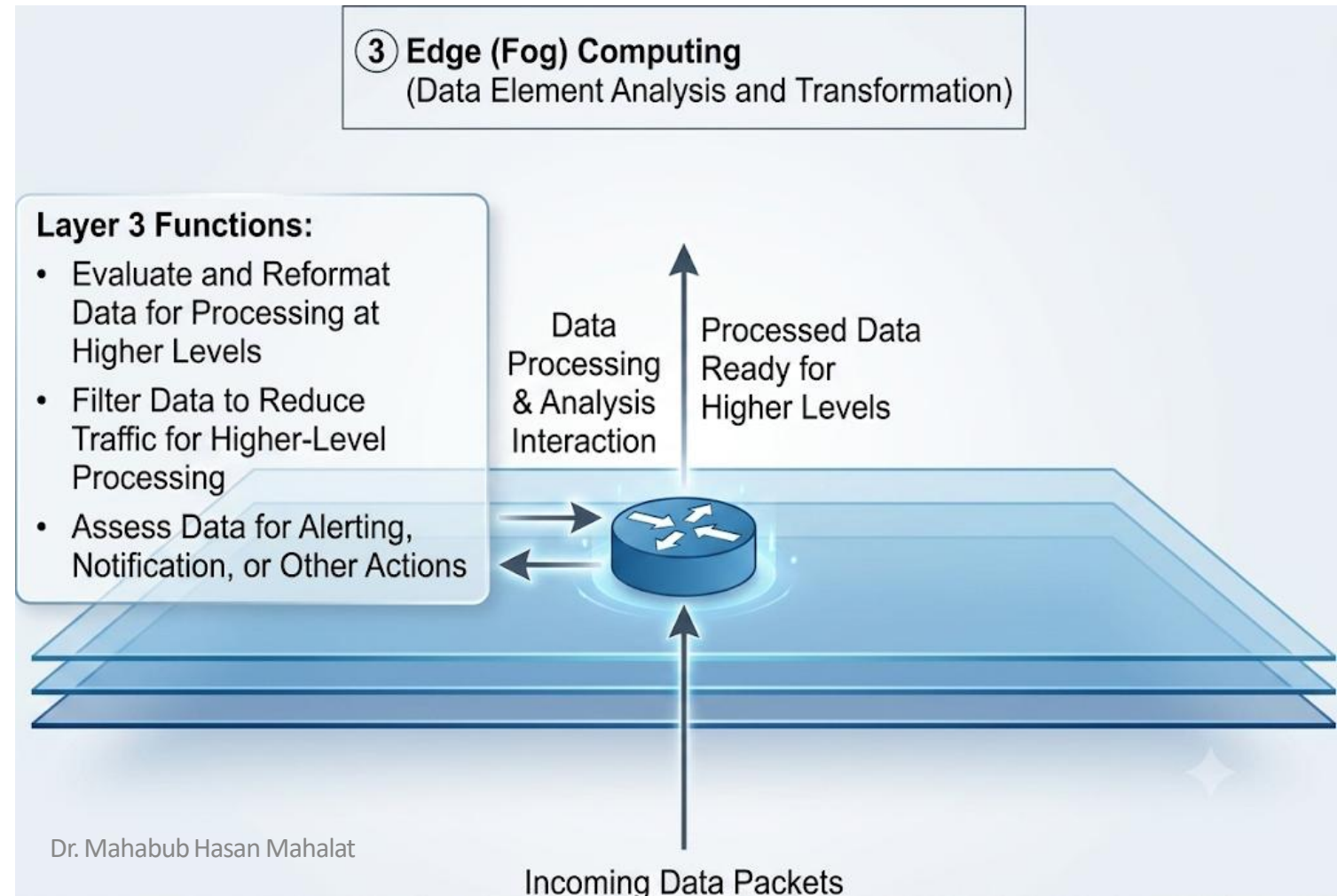
- Some legacy devices are **not IP-enabled** – require communication gateways
- Other devices require **proprietary controllers** for communication
- Over time, **standardization** will increase

Layer 3 – Edge Computing (Fog Computing)

Layer 3: Edge Computing Layer

- Often referred to as the “fog” layer
- Emphasis is on
 - Edge computing involves **information processing at or near the network edge** – as close to the devices as possible
 - **Data reduction** by filtering and cleaning up
 - **Reformatting** and **compressing** data
 - **Initial processing** of data (e.g. alert generation, data validation, etc)

Basic principle: Information processing is initiated **as early** and **as close** to the edge of the network as possible.



Layer 4: Data Accumulation

- It is the layer where streaming (real-time) sensor data is stored permanently in databases or cloud storage for later analysis.
- Captures data and puts it "**at rest**" – converting **data in motion** to **data at rest**.
- **Key Transformation:**
 - **Event-based** data aggregation → **Query-based** processing
 - Real-time networking world → Non-real-time application world

Layer 5 – Data Abstraction

- Focuses on **rendering data and its storage** in ways that enable developing simpler, performance-enhanced applications
- Primary Responsibilities:
 - 1.Reconciling multiple data formats from different sources
 - 2.Assuring consistent semantics of data across sources
 - 3.Confirming data completeness to higher-level applications
 - 4.Consolidating data into one place (ETL, ELT, data replication) or providing access through data virtualization
 - 5.Protecting data with appropriate authentication and authorization
 - 6.Normalizing/denormalizing and indexing data for fast application access

Layer 6 – Application Layer

- This is the layer where **IoT applications** reside – performing business logic and delivering value to end-users
- Interpret data using software applications
- Applications may monitor, control, and provide reports based on analysing the data

Layer 7 – Collaboration and Processes Layer

- This layer addresses how **people and processes** interact with and are empowered by IoT systems
- **Main Purpose**
 - Convert analyzed data into **business value**.
 - Enable **human collaboration** and **automated business processes**.
 - Improve **decision making, productivity, and operational efficiency**.

IT and OT Responsibilities in the IoT Reference Model

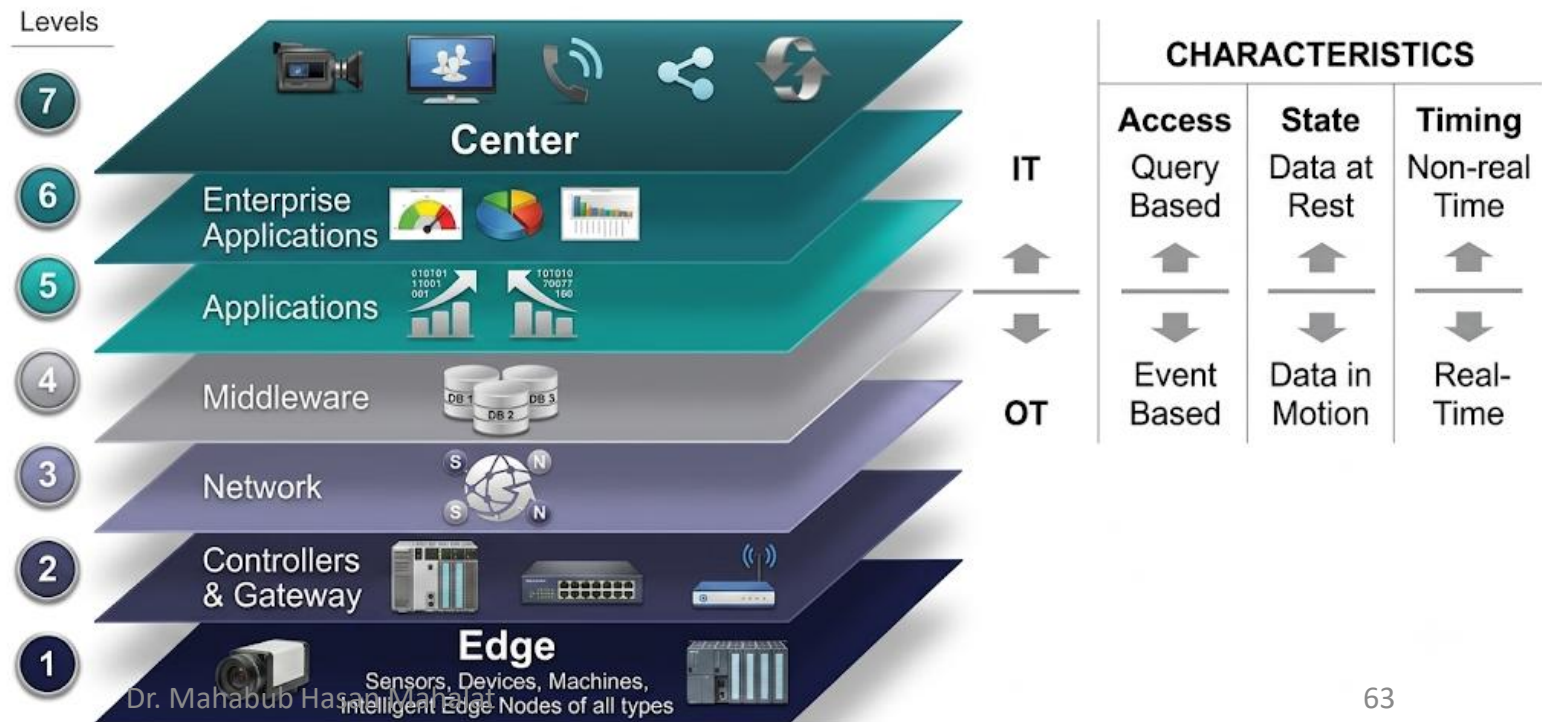
➤ Two Major Domains: OT & IT

➤ Operational Technology (OT)

- Focus on **physical operations** and **real-time control**
- Responsible for Layers 1–3 (Physical Devices, Connectivity, Edge Computing)
- Concerned with: Safety, reliability, low latency, deterministic behavior

➤ Information Technology (IT)

- Focus on **data management**, **analytics**, and **business applications**
 - Responsible for Layers 4–7 (Data Accumulation, Data Abstraction, Application, Collaboration)
 - Concerned with: Scalability, security, data integration, business intelligence
- IoT requires **close collaboration** between IT and OT – a fundamental shift from traditional siloed operations



IoTWF vs oneM2M

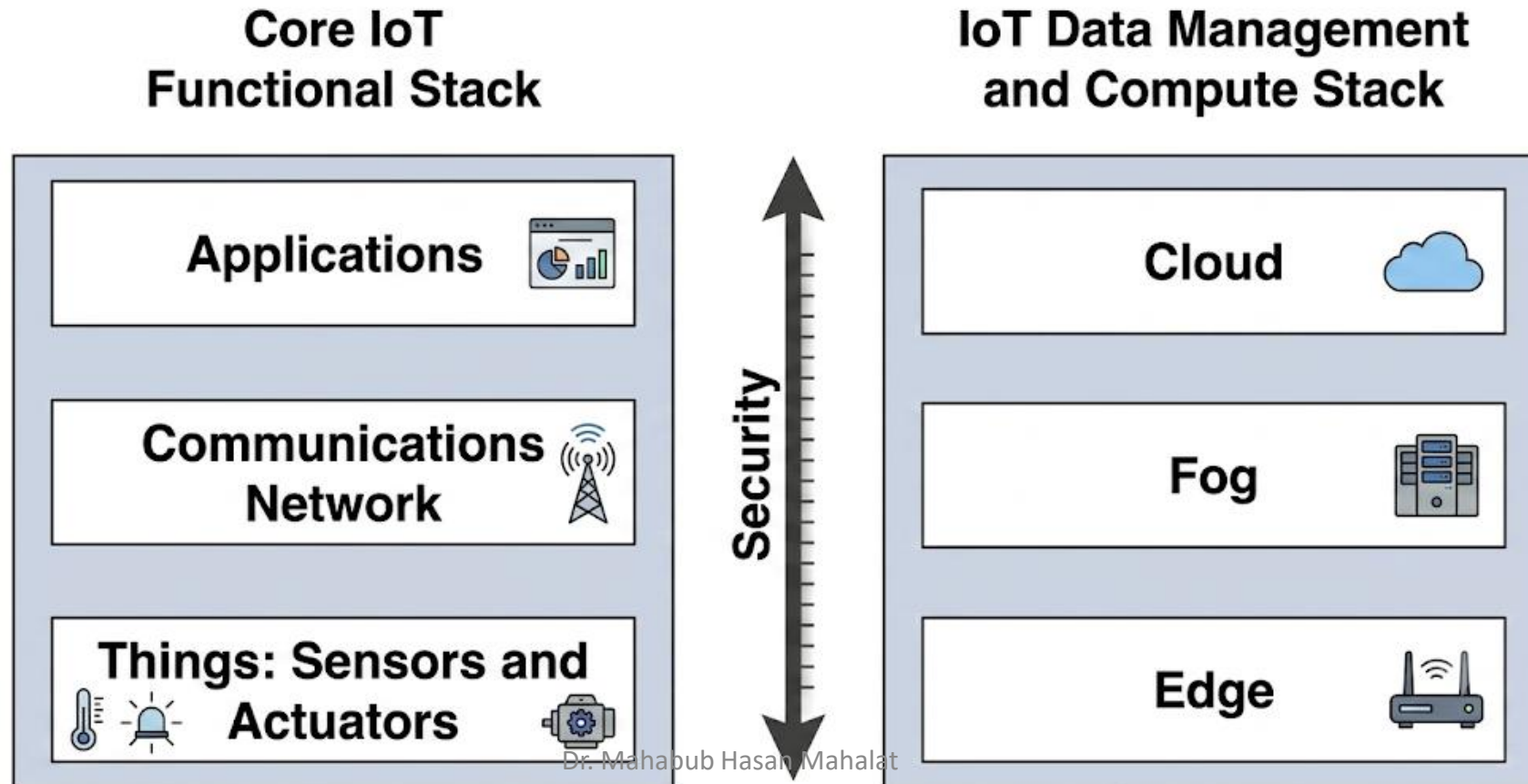
Aspect	IoTWF Architecture	oneM2M Architecture
Origin	IoT World Forum (Cisco-led, 2014)	ETSI + 13 partners (2012)
Number of Layers	7 layers	3 layers (Application, Services, Network)
Focus	End-to-end IoT reference model	Common services layer middleware
Key Concept	Layered decomposition from sensors to people	Horizontal platform with RESTful APIs
Security	Spans all 7 layers	Built into service functions
Approach	Vertical stack with clear layer functions	Horizontal services across verticals

Additional IoT Reference Models

Aspect	Purdue Model	IIRA (IIC)	IoT-A
Origin	Purdue University (1992)	Industrial Internet Consortium (2015)	EU FP7 Lighthouse Project (2010–2013)
Primary Domain	Industrial Control / Manufacturing	Industrial IoT (IIoT)	General IoT (all domains)
Structure	6-level hierarchy (Level 0–5) + DMZ	4 viewpoints + 5 functional domains	4 sub-models (Domain, Information, Functional, Communication)
Key Concept	Strict vertical separation between OT (0-3) and IT (4-5) with DMZ	Stakeholder-driven architecture with multiple viewpoints	Architectural Reference Model (ARM) with reusable design principles
Primary Users	Control engineers, plant operators	Enterprise architects, system integrators, business stakeholders	IoT system designers, researchers, developers
Key Focus	Control hierarchy, network segmentation, ISA-95 compliance	Business transformation, interoperability, safety, security, resilience	Breaking vertical silos, interoperable IoT design, semantic interoperability
Complementary To	ISA-95 (ANSI/ISA-95 Enterprise-Control System Integration)	oneM2M, IoTWF (joint whitepaper published)	oneM2M, FIWARE, ETSI M2M
Strengths	Industry-proven, well understood, clear security boundaries	Vendor-neutral, business-value driven, comprehensive concerns	Conceptual foundation, promotes interoperability, flexible design
Limitations	Originally designed for closed/air-gapped networks – challenges with IIoT cloud integration	Complex to implement; requires significant organizational maturity	Primarily conceptual; implementation requires additional frameworks
Use Cases	Factory automation, SCADA systems, critical infrastructure	Smart manufacturing, predictive maintenance, asset management	Smart cities, e-health, connected vehicles, smart homes

Simplified IoT Architecture

- Simplified framework reduces IoT to its most foundational elements, making it easier to understand key design and deployment principles.
- **Two Parallel Stacks:** Core IoT Functional Stack & IoT Data Management & Compute Stack



Simplified IoT Architecture...

- Simplified framework reduces IoT to its most foundational elements, making it easier to understand key design and deployment principles.
- **Two Parallel Stacks:** Core IoT Functional Stack & IoT Data Management & Compute Stack

Core IoT Functional Stack

Applications Layer – Business logic, analytics, and intelligent controls

Network Communications Layer – Gateways, backhaul, routing, security, QoS

Things Layer – Sensors, actuators, and endpoint devices

IoT Data Management & Compute Stack

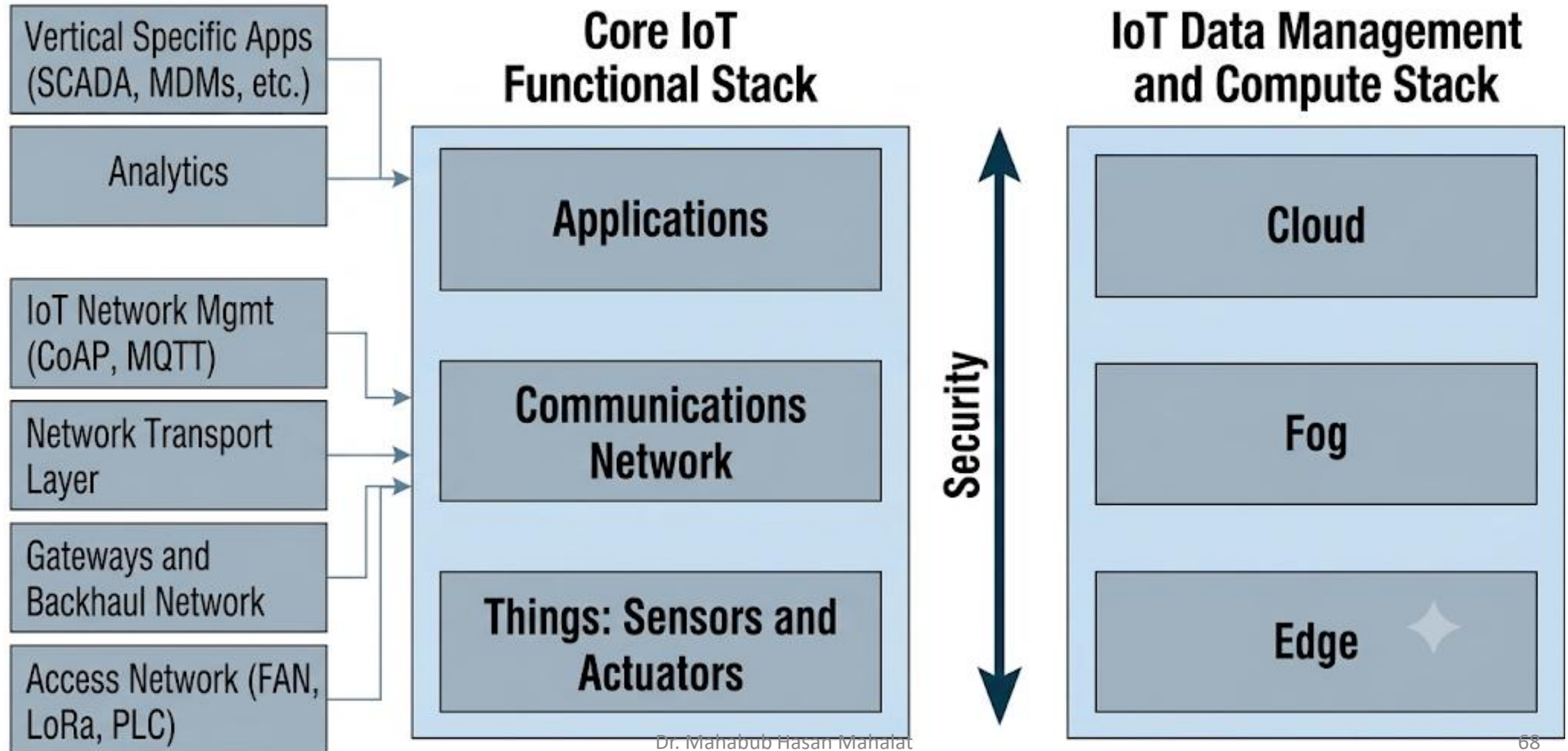
Cloud Layer – Data management in cloud or central data center

Fog Layer – Data management in gateways and transit network

Edge Layer – Data management within the sensors themselves

Expanded View of the Simplified IoT Architecture

EXPANDED VIEW OF IOT ARCHITECTURE



Layer 1 – Things: Sensors and Actuators Layer

- The **Things Layer** is the foundation of any IoT architecture – it consists of the physical "things" (smart objects) that connect to the network, sense the environment, and/or perform physical actions.
- **Sensors** – Detect and measure physical parameters (temperature, motion, humidity, pressure, etc.)
- **Actuators** – Perform physical actions (open/close valves, switch motors, adjust lights, etc.)

Layer 1 – Things: Sensors and Actuators Layer...

- **Architectural Classification of Smart Objects**
- Understanding the diversity of smart objects is essential for selecting appropriate networking protocols, power sources, and data management strategies.
- **Six Key Classification Criteria:**

Classification

Power Source

Mobility

Reporting Frequency

Data Richness

Report Range

Object Density

Key Question

Battery-powered or line-powered?

Mobile or static?

Low or high reporting frequency?

Simple or rich data?

Short or long distance to gateway?

Low or high density per cell?

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Layer 1 – Things: Sensors and Actuators

Layer...

- **Things Layer: Key Takeaway**
- **Foundation of IoT** – All IoT systems begin with the "things" that generate data or perform actions
- **Diversity Drives Architecture** – Variety in smart object types determines protocol and architecture choices
- **Six Key Classifications** – Power, Mobility, Frequency, Richness, Range, Density – each impacts design
- **Trade-Offs are Critical** – Every classification involves trade-offs (battery life vs. range, frequency vs. energy, etc.)
- **Sensors + Actuators** – IoT devices sense the environment (sensors) and act upon it (actuators)

Layer 2 – Communications Network Layer

- IoT vs. IT Network Assets – Key Differences

Aspect	IT Environments	OT/IoT Environments
Physical Environment	Controlled (offices, data centers)	Harsh (deserts, mountains, steel mills, marine)
Temperature Range	Moderate (20–25°C)	Extreme (-40°C to +85°C or more)
Humidity	Controlled	High fluctuations, direct liquid contact possible
Kinetic Forces	Minimal	Shock, constant vibration, sudden acceleration/deceleration
Particulates	Standard dust	High concentrations, caustic materials
Mounting	Rack mount	DIN rail, wall, fence, horizontal mount
Power Supply	Standard AC (110-240V), redundant optional	Wide voltage range, DC power, built-in redundant supplies
Safety Design	Standard	Spark suppression (volatile environments), hazardous location design
Alerting	Software alerts	Physical alarm channels (stack lights, indicator lights)

Physical form factors and operational differences between IT and OT devices are driven by the deployment environment – temperature extremes, humidity, vibration, particulates, and hazardous conditions all impact hardware design and selection

Communications Network Layer: Access Network Sublayer

- Access Network Sublayer physically connects IoT endpoint devices (sensors/actuators) to the network gateway using various wireless or wired technologies.
- Each technology designed for specific use cases → determines frequency band, frame structure, packet size, and communication intervals
- IoT may reuse existing access technologies or use those developed specifically for IoT

Communications Network Layer: Access Network Sublayer

Classification	Range	Technologies
PAN (Personal Area Network)	Few meters	Bluetooth Classic
HAN (Home Area Network)	Few tens of meters	ZigBee, Bluetooth Low Energy (BLE), HomePlug
NAN (Neighborhood Area Network)	Few hundreds of meters	Group of house units
FAN (Field Area Network)	Tens to hundreds of meters	Open outdoor space; group of NANs or HANs
LAN (Local Area Network)	Up to 100 m	Ethernet, IEEE 802.11 (Wi-Fi)
MAN (Metropolitan Area Network)	Up to few kilometers	Various technologies
WAN (Wide Area Network)	> few kilometers	Cellular (3G/4G/5G), LoRa, SigFox, NB-IoT

The choice of access technology is primarily driven by required transmission range, but also by data volume, frequency, power constraints, and mobility.

Communications Network Layer: Gateways and Backhaul Sublayer

- The gateway is responsible for inter-medium communication – forwarding data collected from smart objects via access technologies to the backhaul network, which transports it to the central station for processing.
- **Gateway Placement:**
 - Typically **static** – fixed location connecting sensors to core network
 - **Exception:** DSRC (Dedicated Short-Range Communication) – gateway moves with the vehicle; sensors communicate with mobile gateway via wired/wireless; backhaul uses 5 GHz DSRC for vehicle-to-vehicle and vehicle-to-infrastructure communication

Communications Network Layer: Gateways and Backhaul Sublayer...

- Backhaul Technology Options:

Technology	Range	Throughput	Best For
Ethernet (Wired)	Limited by cable	High	Stable environments (factories, oil/gas fields)
Wi-Fi Mesh	Multiple hops	Halves per hop	Dynamic environments (open mines)
802.11ah (Wi-Fi HaLow)	Up to ~2 km	Moderate	IoT-specific, lower band (<1 GHz), longer range
WiMAX (802.16d)	Up to 50 km	Up to 70 Mbps (short range), 2-3 Mbps (max range)	Licensed bands, reliable backhaul
Cellular (3G/4G/5G)	Similar to WiMAX	Varies	Competitive alternative; vertical/location dependent

The choice between WiMAX, cellular, or other backhaul technologies depends on the **vertical, location, local costs**, and **specific application requirements** – there is no one-size-fits-all solution.

Communications Network Layer: Network Transport Sublayer

- While IoT appears as a **hierarchical structure** (Sensors → Gateway → Headend), real-world implementations involve **multiple communication paths, directions, and models**.
- **Key Drivers of Transport Complexity:**

Driver	Description
Bidirectional Flow	Data flows both upstream (sensor to cloud) and downstream (cloud to sensor)
Multiple Communication Models	Peer-to-peer, point-to-point, point-to-multipoint, unicast, multicast
Dynamic Requirements	On-demand reporting, software updates, load balancing
Cross-System Integration	Multiple utilities (electricity, gas) sharing same pathways
Multiple Media	Power lines, short-range wireless, long-range wireless, wired backhaul

Communications Network Layer: Network Transport Sublayer

- **Real-World Example – Smart Energy Grid:**

Scenario

Basic Meter Reading

High Consumption Alert

Load Balancing (DA)

Home Appliance Integration

Public EV Charging

Software Updates

Communication Change

Vertical push model (meter → gateway → headend) – simple upstream flow

Bidirectional – headend requests on-demand reports from specific meter

Peer-to-peer – meters communicate with neighbors to coordinate AC cycles

Mesh/Partial Mesh – meter talks to washing machine, car charger, home theater

Anywhere connectivity – car logs into central system from any location

Top-down multicast – headend pushes updates to meters and appliances

Communications Network Layer: IoT Network Management Sublayer

- This sublayer manages how data is actually exchanged between smart objects and applications. It sits on top of IP and handles the application-level communication logic.
- **Communication Models (Push vs. Pull):**

Model	Direction	Trigger	Example
Push Model	Sensor → Application	Regular interval or local event (e.g., temperature spike)	Temperature sensor reports every hour automatically
Pull Model	Application → Sensor	On-demand query from server	Headend requests real-time meter reading
Hybrid	Both directions	Combination of scheduled reports + on-demand queries	Most modern IoT systems use this

Communications Network Layer: IoT Network Management Sublayer

- Web-Derived Protocols (Adapted from the Internet):

Protocol	Origin / Base	How it Works	Limitations in IoT
HTTP	Web standard (TCP)	Sensor acts as client, central app as server	"Fat" protocol – too much overhead; not designed for low memory/power/bandwidth
WebSocket	HTML5 standard	Provides simple bidirectional connection over a single TCP pipe	Often just a tunnel; usually combined with other protocols (like MQTT)
XMPP	Instant Messaging (TCP)	Supports presence, contact lists, and publish/subscribe	Relies on TCP – forces open sessions, consuming memory on constrained devices

Layer 3 – Applications and Analytics Layer

- Once smart objects are connected to a network, the true power of IoT emerges in the applications **that collect, analyze, and act** upon the data generated by these objects.
- **Two Main Application Types:**

Application Type	Function	Example
Analytics Application	Collects data from multiple sensors, processes it, and displays insights (trends, statistics, system states)	Dashboard showing storm likelihood based on thousands of sensors
Control Application	Controls behavior of smart objects or related systems based on analyzed data	Pressure sensor increases pump speed when pressure drops

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Analytics – Data vs. Network Analytics

➤ Two Types of Analytics in IoT: **Data Analytics & Network Analytics**

➤ **Data Analytics**

Aspect	Description
Purpose	Process sensor data to provide intelligent insights about the IoT system
Basic Example	Dashboard shows alarm when weight sensor detects empty shelf
Complex Example	Combine temperature, pressure, wind, humidity to predict storm path
Self-Monitoring	Robot reports movement degradation → alerts for maintenance before part breaks

Analytics – Data vs. Network Analytics...

➤ Two Types of Analytics in IoT: **Data Analytics & Network Analytics**

➤ **Network Analytics**

Aspect	Description
Purpose	Monitor connectivity health of the IoT network itself
Why Important	Loss of connectivity stops data flow → analytics platform fails → control module cannot act
Real-World Impact	Automated dump trucks in open mines stop upon connectivity loss to prevent accidents
Depth of Analysis	Basic = occasional status view; Advanced = near-real-time performance trending

IoT Data Management and Compute Stack – The Data Challenge

➤ The Scale Problem:

Metric	Traditional IT	IoT
Connected Devices	Thousands	Millions to Billions (50B+ by 2020)
Data Generated	Structured, application-driven	Unstructured, continuous streams
Data Value	Most data is useful	Majority is low-value (polling, keep-alive)

- The massive scale of IoT networks creates a data volume crisis – traditional IT networks and data centers are not designed to handle.
- Most IoT data (polling messages, keep-alive signals) has **little individual value** – but when analyzed in aggregate (e.g., millions of meters losing connection simultaneously), it provides critical insights about system-wide event.

The Problem – Why Centralized Cloud Processing Fails

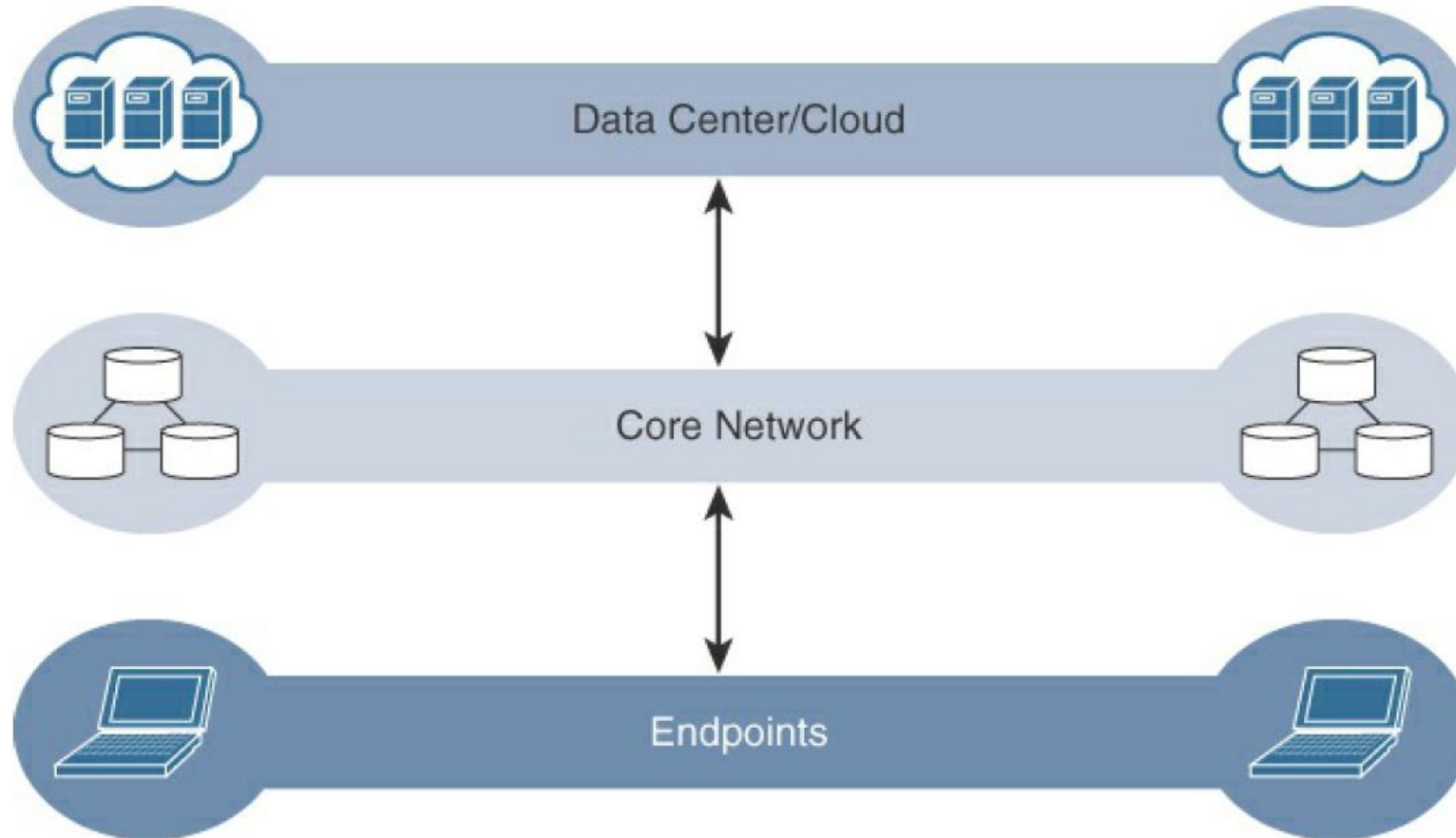


Fig. Traditional IT Cloud Computing Model
Simple – one central application has visibility over all nodes.

The Problem – Why Centralized Cloud Processing Fails...

➤ Limitations of Centralized Cloud Processing:

Challenge	Description	Example
Latency	Milliseconds matter for industrial control	Preventing manufacturing line shutdowns, restoring electrical service
Bandwidth	Not practical to transport vast data to cloud	Offshore oil rig: 500 GB/week; Commercial jet: 10 TB/30 min flight
Local Efficiency	Conditions vary by location – cloud analysis may be unnecessary	Environmental response differs by site; local decisions needed
Impedance Mismatch	Network/application cannot ingest/store/analyze incoming data volume	1 million meters = 1 TB data/day
Backhaul Constraints	Unreliable, expensive (3G/LTE/satellite), per-byte costs	Limited bandwidth, high latency (hundreds to thousands of ms)

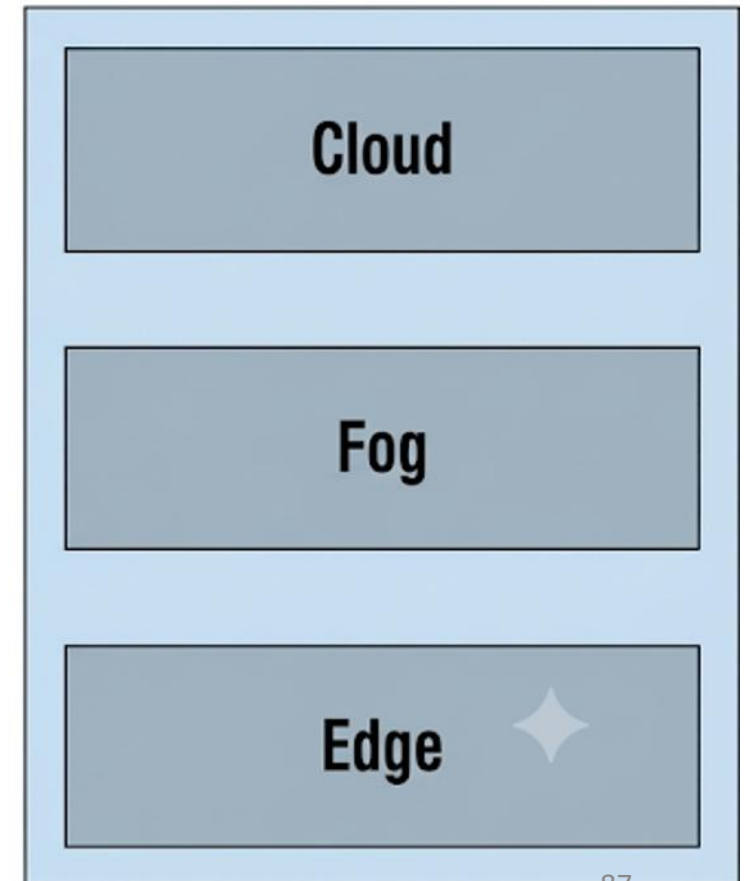
The Solution – Tiered Data Management (Edge → Fog → Cloud)

- IoT systems need a tiered approach to data management – processing data at multiple points throughout the architecture rather than sending everything to the cloud.

- **Three-Layer Data Management Stack:**

Layer	Location	Function
Edge Layer	Within sensors/devices	Initial data filtering, local decisions
Fog Layer	Gateways, routers, local servers	Aggregation, temporary storage, local analytics
Cloud Layer	Central data center / Cloud	Long-term storage, complex analytics, business intelligence

IoT Data Management and Compute Stack

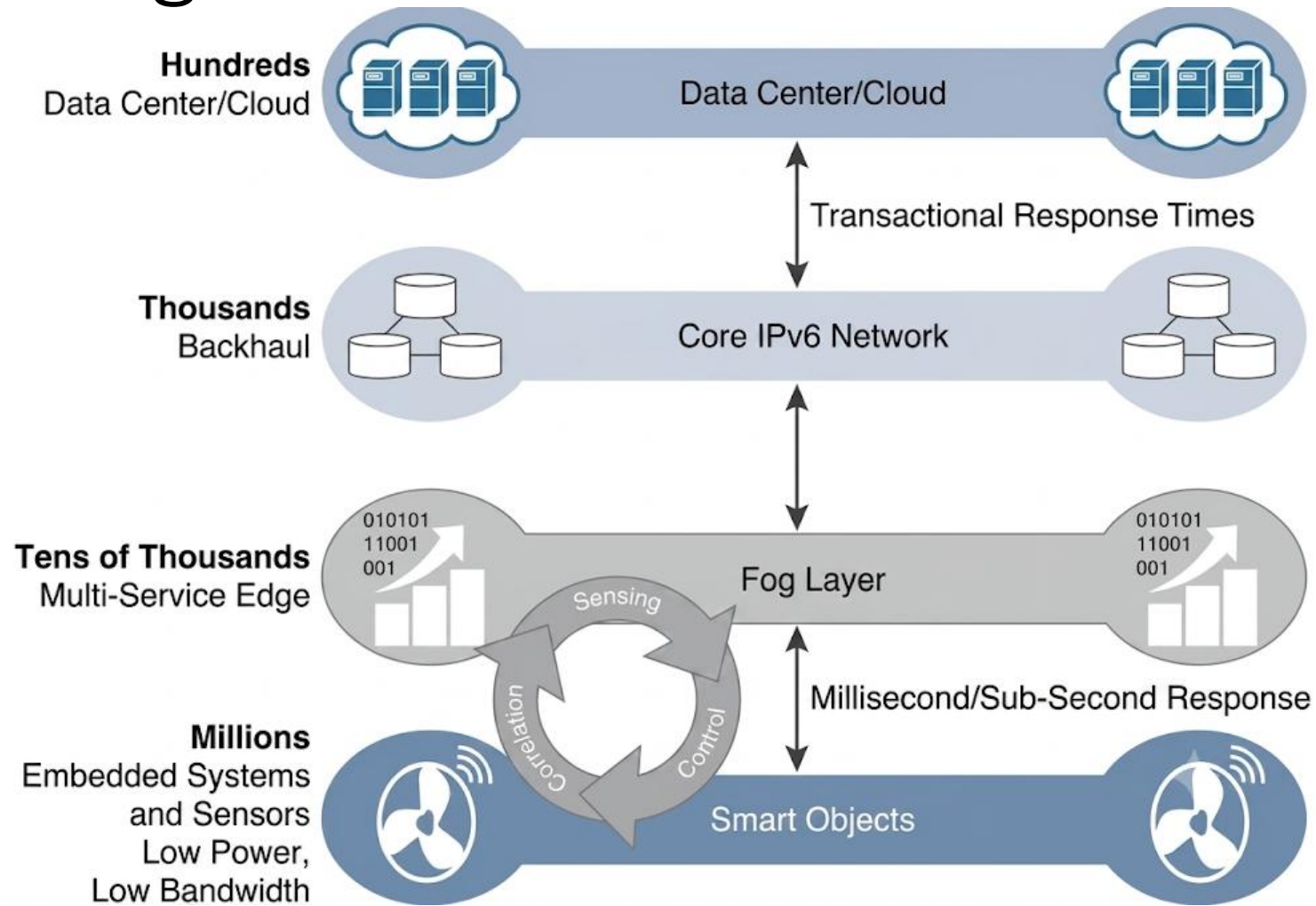


The Solution – Tiered Data Management (Edge → Fog → Cloud)...

➤ Why Tiered Processing is Essential:

Requirement	How Tiered Approach Helps
Minimize Latency	Process critical data at edge/fog – immediate action
Conserve Bandwidth	Filter/discard low-value data before sending to cloud
Reduce Cost	Less backhaul usage, lower data transfer costs
Enable Scale	Distributed processing handles billions of devices
Local Efficiency	Site-specific decisions made locally without cloud dependency

Fog Computing



- Any device with computing, storage, and network connectivity can be a fog node – including industrial controllers, switches, routers, embedded servers, and IoT gateways.

Why Fog Computing?

Challenge	How Fog Solves It
Latency	Analyzes data close to source → milliseconds response (vs. cloud seconds)
Bandwidth	Filters/aggregates data → sends only relevant info upstream
Cost	Reduces expensive backhaul usage (3G/LTE/satellite)
Security	Keeps sensitive data inside local network
Contextual Awareness	Geographic proximity provides better understanding of local conditions

Example: Oil derrick fog node monitors all sensors locally – analyzes context, sends only relevant alerts over backhaul, reducing upstream traffic and enabling real-time response.

Fog Computing – Defining Characteristics and Applications

Five Defining Characteristics:

Characteristic	Description
1. Contextual Location Awareness & Low Latency	Sits as close to IoT endpoints as possible – delivers distributed computing with millisecond response
2. Geographic Distribution	Decentralized – in contrast to centralized cloud; deployments are widely distributed
3. Deployment Near IoT Endpoints	Typically deployed with large numbers of endpoints (e.g., 3000–4000 nodes per gateway router)
4. Wireless Communication	Wireless access enables massive scale connectivity between fog nodes and endpoints
5. Real-Time Interactions	Supports real-time processing (not batch) – preprocessing enables upper-layer batch analytics

Fog Applications – Real-World Examples

Domain	Fog Application
Mining	Tire pressure sensors on truck → fog node monitors all sensors → sends alerts only when problem detected
Smart Grid	Monitor voltage, power quality, outages → local response without waiting for cloud
Transportation	Train braking systems → real-time response based on sensor data
Security	Video camera zoom → triggered by local motion detection
Industrial	Open/close valve based on pressure reading → immediate action
Smart Home	Lock doors, change equipment settings based on presence detection

Edge Computing

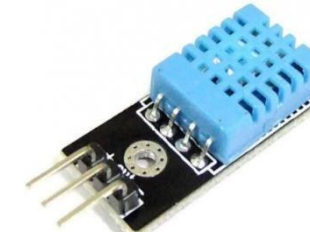
- Edge computing pushes IoT computing even further than fog – directly into the sensors and IoT devices themselves. It is also sometimes called "mist" computing (**cloud → fog → mist**).
- **Why Edge Computing?**

Benefit	Description
Ultra-Low Latency	Decisions made at the device – milliseconds faster than fog
Bandwidth Savings	Filter data before it even leaves the sensor
Localized Decisions	Each device monitors its own immediate environment
Resilience	Continues operating even if network/fog is unavailable

Sensors, Actuators, and Smart Objects in IoT

➤ Sensors

- Measure physical or environmental conditions.
- Convert real-world data into digital signals.
- **Examples:** Temperature, humidity, motion, light, gas.



Sensor

➤ Actuators

- Perform physical actions based on received commands.
- Convert electrical signals into mechanical or physical output.

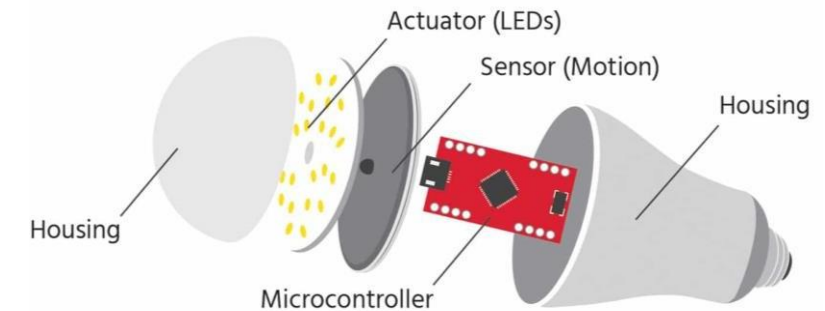


Actuator

➤ **Examples:** Relay, motor, LED, buzzer, valve.

➤ Smart Objects

- A **smart object** is a physical device that integrates:
- **Embedded system** (Microcontroller, memory, power supply)
- **Sensors** (to collect data)
- **Actuators** (to perform actions)
- **Communication module** (Wi-Fi, Bluetooth, ZigBee, LoRa, etc.)
- **Example:** Smart Street Light



Smart Object

- Motion sensor detects a pedestrian. -> Microcontroller processes the input. -> LED light (actuator) turns ON -> Status is sent to the cloud or monitoring system.

Sensors: Classification and Characteristics

- A sensor is a transducer that detects a physical phenomenon and converts it into an electrical signal for monitoring, control, or decision-making in IoT systems.
- Sensors can be classified in various ways:

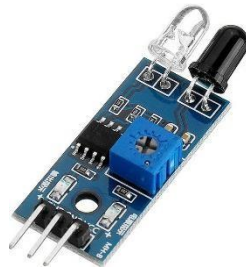
Classification	Types	Examples
Based on Energy Source	Active, Passive	Radar, LDR
Based on Contact	Contact, Non-contact	Thermocouple, IR Sensor
Based on Output	Analog, Digital	LM35, DHT22
Based on Measurement	Absolute, Relative	Absolute Pressure, Differential Pressure
Based on Sensing Mechanism	Optical, Resistive, Capacitive, Piezoelectric, Electromagnetic, Electrochemical	LDR, Strain Gauge, Capacitive Humidity, Piezo Sensor

Sensor Types: What it measures

Sensor Type	Description	Example
Position	➤ Measures the position of an object ➤ Position could be absolute/relative ➤ Position sensor could be linear, angular, or multi-axis	➤ Proximity sensor ➤ Potentiometer ➤ Inclinator
Occupancy	➤ Detects the presence of people and animals in a surveillance area ➤ Generates signal even when a person is stationary	➤ Radar Sensor
Motion	➤ Detects the movement of people and objects	➤ Passive Infrared (PIR) Sensor



Ultrasonic Proximity Sensor



Infrared Proximity Sensor



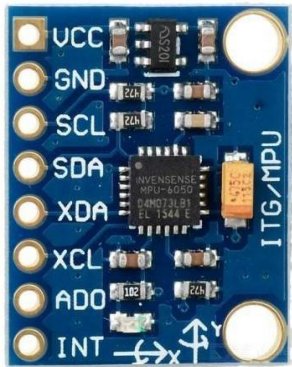
Microwave Radar Sensor



PIR Motion Sensor

Sensor Types: What it measures...

Sensor Type	Description	Example
Velocity and Acceleration	➤ Velocity sensor measures how fast an object moves ➤ Acceleration sensor measures the changes in velocity	➤ Gyroscope ➤ Accelerometer
Force	➤ Detects whether a physical force is applied and the magnitude of the force	➤ Tactile sensor ➤ Viscometer
Pressure	➤ Measuring the force applied by liquids or gases ➤ It is measured as force per unit area	➤ Barometer ➤ Piezometer



Gyroscope



Capacitive Touch Sensor



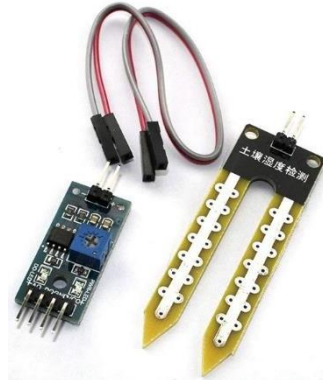
Barometric Pressure Sensor

Sensor Types: What it measures...

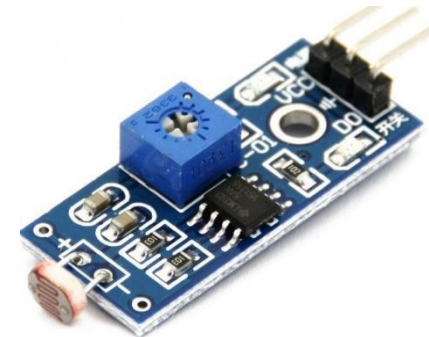
Sensor Type	Description	Example
Flow	➤ Detects the rate of fluid flow through a system in given period of time	➤ Water meter ➤ Anemometer
Humidity	➤ Detects amount of water vapour in the air ➤ Can be measured in absolute/relative scale	➤ Hygrometer ➤ Soil moisture sensor
Light	➤ Detects the presence of light	➤ LDR light sensor ➤ Photodetector ➤ Flame Sensor



Water meter



Soil moisture sensor



LDR light sensor



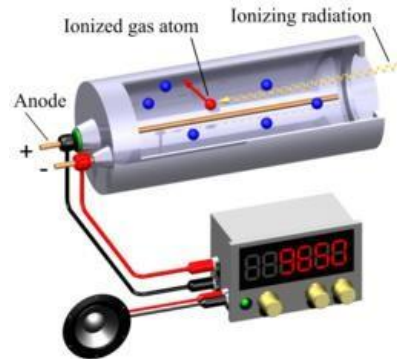
Flame sensor

Sensor Types: What it measures...

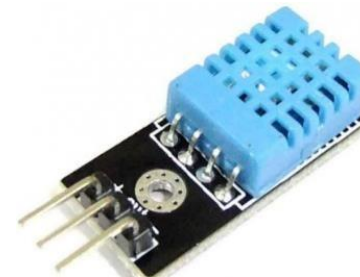
Sensor Type	Description	Example
Radiation	➤ Detects the radiation in the environment	➤ Neutron detector ➤ Geiger-Muller counter
Temperature	➤ Measures the amount of heat or cold present in the system ➤ Two type: contact / non-contact	➤ Thermometer ➤ Temperature gauge ➤ Calorimeter



Neutron detector



Geiger-Muller counter



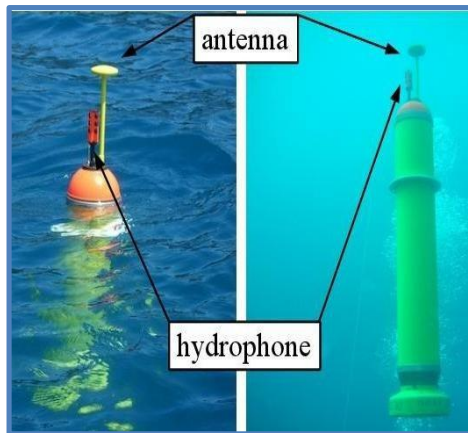
Temperature
Sensor



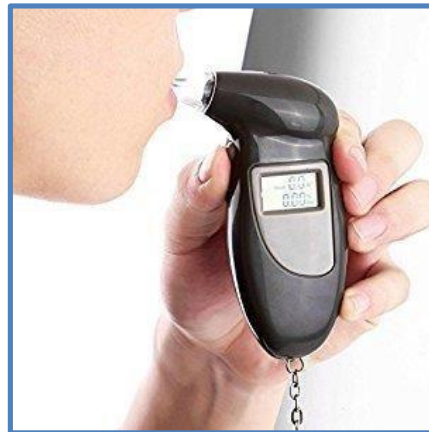
Thermo-
Hygrometer

Sensor Types: What it measures...

Sensor Type	Description	Example
Acoustic	➤ Measures sound level	➤ Microphone ➤ Hydrophone
Chemical	➤ Measures the concentration of a chemical (e.g. CO ₂) in a system	➤ Smoke detector ➤ Breathalyzer
Biosensor	➤ Detects various biological elements, such as organisms, tissues, cells, enzymes, antibodies, nucleic acid, etc.	➤ Pulse oximeter ➤ Electrocardiograph (ECG) ➤ Blood glucose biosensor



Hydrophone



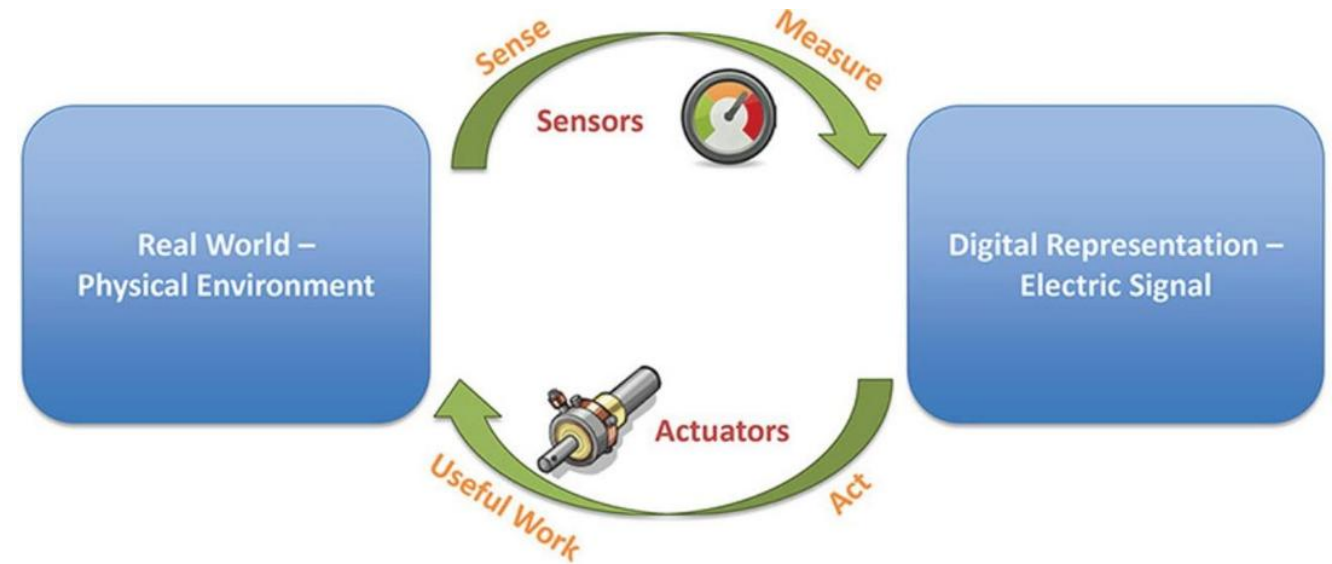
Breathalyzer



Pulse oximeter

Actuators

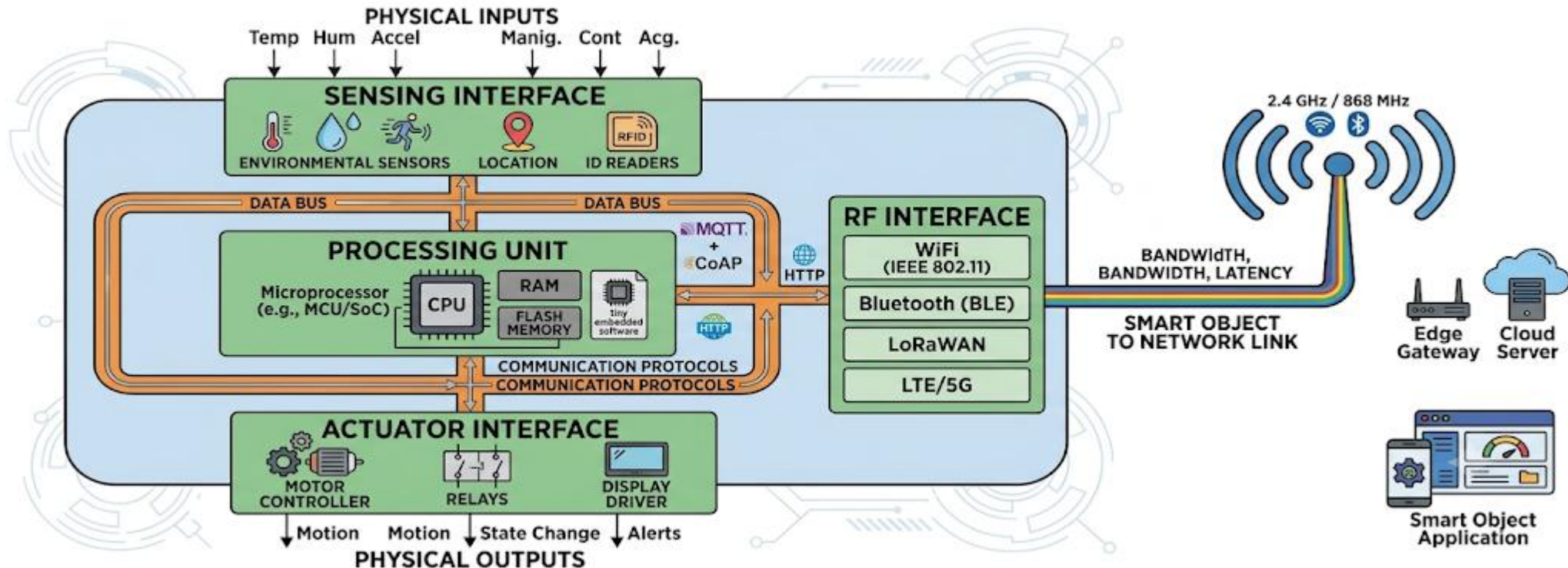
- An actuator is a device that receives a control signal from a controller and converts electrical, hydraulic, pneumatic, or thermal energy into a physical action.
- Common Ways to Classify Actuators



Classification	Types	Examples
Based on Energy Source	Electrical, Hydraulic, Pneumatic, Thermal	DC Motor, Hydraulic Cylinder, Pneumatic Valve
Based on Motion	Linear, Rotary	Linear Actuator, Servo Motor
Based on Output	Binary (ON/OFF), Continuous	Relay, Servo Motor
Based on Power	Low, Medium, High	LED, Servo, Industrial Motor
Based on Application	Industrial, Medical, Automotive, Smart Home	Robotic Arm, Insulin Pump, Smart Lock

Smart Objects

- A smart object is a physical device embedded with sensing, processing, communication, and actuation capabilities that can interact with its environment and exchange data over a network.



Characteristics and Examples of Smart Objects

➤ Characteristics

- **Context-aware** – Detects changes in its environment.
- **Connected** – Communicates with other devices and cloud services.
- **Autonomous** – Makes local decisions based on embedded logic.
- **Programmable** – Firmware can be updated remotely (OTA).
- **Energy-efficient** – Optimized for low-power operation.
- **Secure** – Supports authentication, encryption, and secure boot.

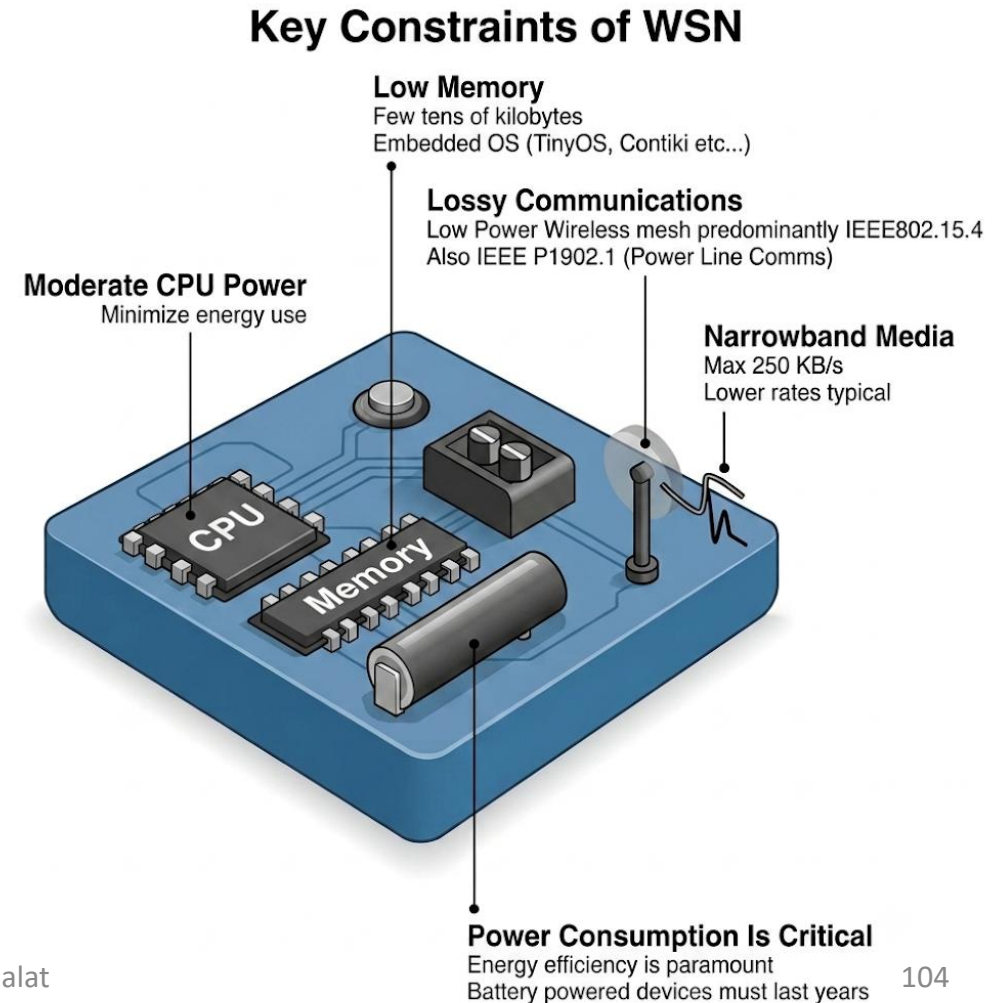
➤ Examples:

Smart Object	Sensors	Actuators	Communication
Smart Thermostat	Temperature	HVAC Control	Wi-Fi
Smart Door Lock	Fingerprint, RFID	Lock Motor	BLE, Wi-Fi
Smart Irrigation Node	Soil Moisture	Water Pump	LoRa, Wi-Fi
Wearable Device	Heart Rate, Accelerometer	Vibration Motor	Bluetooth
Smart Street Light	Motion, Light	LED Lamp	LoRa, NB-IoT

Wireless Sensor Networks (WSNs)

- WSNs are made up of wirelessly connected smart objects, sometimes referred to as **nodes**.
- They require no fixed infrastructure – offering powerful flexibility for deployments.
- Key Constraints of WSN

Limitation	Impact
Limited Processing Power	Cannot perform complex computations locally
Limited Memory	Restricted storage for code and data
Lossy Communication	Unreliable wireless links – packet loss is common
Limited Transmission Speeds	Low data rates – cannot stream high-bandwidth data
Limited Power	Battery-operated – energy conservation is critical



WSN – Communication Patterns and Data Aggregation

Pattern	Description	Example
Event-Driven	Transmission triggered only when specific event or threshold is detected	Blood pressure alert when critically high/low reading is measured
Periodic	Transmission occurs at fixed time intervals	Postoperative vitals (temperature, BP) sent every hour

The choice depends on the specific application requirements (criticality, power constraints, data value).



End of Unit 1

